

AUTOMORPHIC FORMS

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1. MODULAR FORMS

Definition 1.0.1. (♠ TODO: define congruence subgroups for more general arithmetic groups and describe how these are instances of those) Let N be a positive integer.

- The *principal congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of level N* is the group $\Gamma(N)$ defined by

$$\Gamma(N) := \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) : \gamma \equiv I_2 \pmod{N}\}.$$

- The subgroup $\Gamma_1(N)$ of $\mathrm{SL}_2(\mathbb{Z})$ is defined by

$$\Gamma_1(N) := \left\{ \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : a \equiv d \equiv 1 \pmod{N}, c \equiv 0 \pmod{N} \right\}.$$

- The subgroup $\Gamma_0(N)$ of $\mathrm{SL}_2(\mathbb{Z})$ is defined by

$$\Gamma_0(N) := \left\{ \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{N} \right\}.$$

- A *congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$* is any subgroup $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ that contains $\Gamma(N)$ for some $N \geq 1$.

Note that

$$\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N) \subset \mathrm{SL}_2(\mathbb{Z}).$$

Moreover, the natural reduction homomorphism

$$\mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$$

is surjective with kernel $\Gamma(N)$. Therefore, we have an isomorphism

$$\mathrm{SL}_2(\mathbb{Z})/\Gamma(N) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}).$$

In particular, $\Gamma(N)$ and hence $\Gamma_1(N)$ and $\Gamma_0(N)$ are all of finite index in $\mathrm{SL}_2(\mathbb{Z})$.

Definition 1.0.2 (Principal congruence subgroup $\Gamma(N)$ of $\mathrm{SL}_2(\mathbb{Z})$). Let N be a positive integer. The *principal congruence subgroup of level N* , denoted

$$\Gamma(N),$$

is the kernel of the reduction homomorphism $\mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. Explicitly,

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \mid a \equiv d \equiv 1 \pmod{N}, b \equiv c \equiv 0 \pmod{N} \right\}.$$

Equivalently, $\Gamma(N) = \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) \mid \gamma \equiv I_2 \pmod{N}\}$, where I_2 is the identity matrix in $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$.

The subgroup $\Gamma(N)$ is a normal subgroup of $\mathrm{SL}_2(\mathbb{Z})$, and the natural reduction homomorphism

$$\mathrm{SL}_2(\mathbb{Z}) \longrightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$$

is a surjective group homomorphism with kernel $\Gamma(N)$. Therefore, there is an isomorphism

$$\mathrm{SL}_2(\mathbb{Z})/\Gamma(N) \cong \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}).$$

In particular, $\Gamma(N)$ has finite index in $\mathrm{SL}_2(\mathbb{Z})$, with index equal to $|\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})|$.

(♠ TODO: modular curve) The principal congruence subgroup $\Gamma(N)$ is used in the construction of the modular curve $X(N)$.

Lemma 1.0.3. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ (Definition 1.0.1).

There is some integer $h \in \mathbb{Z}_{>0}$ such that $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix} \in \Gamma$.

Definition 1.0.4 (Upper half-plane). Let $\mathbb{C}$ denote the field of complex numbers. The *upper half-plane in the complex plane* is the subset

$$\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$$

where $\mathrm{Im}(z)$ denotes the imaginary part of z .

The upper half-plane is an open subset of the topological space $\mathbb{C}$, a convex set, and a simply connected subset of $\mathbb{C}$. It carries a natural structure of a complex analytic manifold of dimension 1 over $\mathbb{C}$.

Definition 1.0.5 (Extended upper half-plane). Let $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ be the upper half-plane in the complex plane (Definition 1.0.4). The *extended upper half-plane* is the set

$$\widehat{\mathbb{H}} = \mathbb{H} \cup \mathbb{R} \cup \{\infty\}$$

where $\mathbb{R}$ is the real line and ∞ is a point at infinity. Equivalently, $\widehat{\mathbb{H}} = \mathbb{H} \cup \mathbb{P}^1(\mathbb{R})$, identifying $\mathbb{R} \cup \{\infty\}$ with $\mathbb{P}^1(\mathbb{R})$.

(♠ TODO: riemann sphere) The extended upper half-plane carries the subspace topology inherited from the Riemann sphere $\mathbb{C} \cup \{\infty\}$, where a basis of neighborhoods is given by: for $x \in \mathbb{R}$, open semicircles centered at x lying in $\mathbb{H}$ together with open intervals on $\mathbb{R}$ containing x ; and for ∞ , complements in $\widehat{\mathbb{H}}$ of compact subsets of $\mathbb{H}$. With this topology, $\widehat{\mathbb{H}}$ is homeomorphic to the closed unit disk and serves as the domain on which the modular action (Definition 1.0.8) of subgroups of $\text{SL}_2(\mathbb{Z})$ extends from $\mathbb{H}$ alone to include the cusps (Definition 1.2.3) $\mathbb{R} \cup \{\infty\}$.

Definition 1.0.6 (Closed upper half-plane). Let $\mathbb{C}$ denote the field of complex numbers. The *closed upper half-plane in the complex plane* is the subset

$$\overline{\mathbb{H}} = \{z \in \mathbb{C} : \text{Im}(z) \geq 0\}$$

where $\text{Im}(z)$ denotes the imaginary part of z .

The closed upper half-plane is the closure of the upper half-plane in the complex plane (Definition 1.0.4) in $\mathbb{C}$, and its boundary $\overline{\mathbb{H}} \setminus \mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) = 0\}$ is canonically identified with the field of real numbers.

Definition 1.0.7 (Nontangential limit of a holomorphic function on the upper half-plane).

(♠ TODO: upper half plane) Let $\mathbb{C}$ denote the field of complex numbers, and let $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ be the upper half-plane. Let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a holomorphic function.

For a point $x \in \mathbb{R}$ and $\alpha > 1$, the *nontangential approach region/Stolz angle at x with aperture α* is the set

$$\Gamma_\alpha(x) = \{z \in \mathbb{H} : |z - x| < \alpha \cdot \text{Im}(z)\}.$$

We say that f has the *nontangential limit* $L \in \mathbb{C}$ at $x \in \mathbb{R}$ if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$|f(z) - L| < \epsilon$$

whenever $z \in \Gamma_\alpha(x)$ and $|z - x| < \delta$, for some (equivalently, any) fixed $\alpha > 1$.

Definition 1.0.8 (Modular group action). Let $\Gamma \subseteq \text{SL}_2(\mathbb{Z})$ be a subgroup. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and $\tau \in \mathbb{H} := \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$, define the action

$$\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}.$$

This is called the *modular action of Γ on the upper half-plane $\mathbb{H}$* .

Definition 1.0.9 (Modular action of a subgroup of $\mathrm{SL}_2(\mathbb{Z})$ on the extended upper half-plane). Let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ be a subgroup. The *modular action of Γ on the extended upper half-plane* $\widehat{\mathbb{H}}$ extends the modular action of Γ on $\mathbb{H}$ (Definition 1.0.8) to all of $\widehat{\mathbb{H}} = \mathbb{H} \cup \mathbb{R} \cup \{\infty\}$ (Definition 1.0.5) as follows. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$:

1. If $\tau \in \mathbb{H}$, then $\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}$ (the original modular action).
2. If $\tau \in \mathbb{R}$ and $c\tau + d \neq 0$, then $\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d} \in \mathbb{R}$.
3. If $\tau \in \mathbb{R}$ and $c\tau + d = 0$, then $\gamma \cdot \tau := \infty$.
4. $\gamma \cdot \infty := \begin{cases} a/c & \text{if } c \neq 0, \\ \infty & \text{if } c = 0. \end{cases}$

This action is well-defined and gives a group homomorphism $\Gamma \rightarrow \mathrm{Bij}(\widehat{\mathbb{H}})$. Since each γ acts as a Möbius transformation on $\mathbb{P}^1(\mathbb{C})$, the restriction to $\widehat{\mathbb{H}}$ preserves $\mathbb{H}$, maps $\mathbb{R} \cup \{\infty\}$ to itself, and is continuous with respect to the natural topology on $\widehat{\mathbb{H}}$.

Definition 1.0.10 (Weight- k slash operator). Let $k \in \mathbb{Z}$. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ and any function $f : \mathbb{H} \rightarrow \mathbb{C}$, the *weight- k slash operator* (or simply the *weight- k operator*) is defined as

$$(f|_k\gamma)(\tau) := (c\tau + d)^{-k} f\left(\frac{a\tau + b}{c\tau + d}\right).$$

$f|_k\gamma$ is also often denoted by $f[\gamma]_k$.

Definition 1.0.11 (Weakly modular form). Let $k \in \mathbb{Z}$, and let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ be a subgroup. A meromorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ is called a *weakly modular form of weight k with respect to Γ* or is said to *satisfy the modular transformation property of weight k with respect to Γ* if

$$f|_k\gamma = f \quad \text{for all } \gamma \in \Gamma,$$

(Definition 1.0.10) i.e.

$$f(z) = (cz + d)^{-k} f\left(\frac{az + b}{cz + d}\right)$$

for all $z \in \mathbb{H}$.

Lemma 1.0.12. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ (Definition 1.0.1). Let $h \in \mathbb{Z}_{>0}$ be such that $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix} \in \Gamma$ by Lemma 1.0.3. Every weakly modular form f (Definition 1.0.11) of some weight $k \in \mathbb{Z}$ is $h\mathbb{Z}$ -periodic, i.e. $f(z + h) = f(z)$ for all $z \in \mathbb{H}$ (Definition 1.0.4).

In particular, there exists a function $g : D' \rightarrow \mathbb{C}$ where $D' \subset \mathbb{C}$ is the punctured unit disk such that $f(z) = g(q_h)$ where $q_h = e^{2\pi iz/h}$.

Definition 1.0.13. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ (Definition 1.0.1). Let $h \in \mathbb{Z}_{>0}$ be minimal such that $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$ by Lemma 1.0.3.

Let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a meromorphic function of the form $f(z) = g(q_h)$ for some meromorphic function $g : D' \rightarrow \mathbb{C}$ on the punctured unit disk D' where $q_h = e^{2\pi iz/h}$ (e.g. weakly modular forms (Definition 1.0.11) satisfy this condition, see Lemma 1.0.12). In particular, g has a Laurent expansion near $q_h = 0$ (which corresponds to $\mathrm{Im}(z) \rightarrow \infty$) (♠ TODO: cite this as a complex analytic fact) and hence f has a Fourier expansion:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n q_h^n.$$

The values $a_n \in \mathbb{C}$ are called the *Fourier coefficients of f* .

We say that f is *meromorphic at ∞* if g is meromorphic at $q_h = 0$, i.e. the Fourier expansion of f has the form

$$f(z) = \sum_{n=n_0}^{\infty} a_n q_h^n.$$

for some $n_0 \in \mathbb{Z}$. We say that f is *holomorphic at ∞* if g is holomorphic at $q_h = 0$, i.e. the Fourier expansion of f has the form

$$f(z) = \sum_{n=0}^{\infty} a_n q_h^n.$$

If f is holomorphic at ∞ and we additionally have $a_0 = 0$, then we say that f has a *cusp at ∞* .

Definition 1.0.14 (Fourier expansion of a periodic holomorphic function on the upper half-plane). (♠ TODO: AI generated, verify) Let $h \in \mathbb{R}_{>0}$, let $\mathbb{H}$ denote the upper half-plane (Definition 1.0.4), and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a holomorphic function satisfying

$$f(z + h) = f(z) \quad \text{for all } z \in \mathbb{H}.$$

For each integer $n \in \mathbb{Z}$, the *n -th Fourier coefficient of f* is the complex number

$$a_n = \frac{1}{h} \int_{z_0}^{z_0+h} f(z) e^{-2\pi i n z/h} dz,$$

where the integral is taken along any piecewise continuously differentiable path in $\mathbb{H}$ from z_0 to $z_0 + h$ for some $z_0 \in \mathbb{H}$ (the value is independent of the choice of path and of z_0 by the Cauchy integral theorem).

The *Fourier expansion (or Fourier series) of f* is the formal series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n z/h},$$

where a_n are the Fourier coefficients of f . (Theorem 1.0.15) (♠ TODO: why the $\sim$, why not just an equality?)

Theorem 1.0.15 (Periodic holomorphic function on the upper half-plane descends to the punctured disk and has a Fourier–Laurent expansion). (♠ TODO: AI generated, verify) Let $h \in \mathbb{R}_{>0}$, let $\mathbb{H}$ denote the upper half-plane (Definition 1.0.4), and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a holomorphic function satisfying

$$f(z + h) = f(z) \quad \text{for all } z \in \mathbb{H}.$$

Define the map $\phi : \mathbb{H} \rightarrow \mathbb{C}$ by

$$\phi(z) = e^{2\pi iz/h},$$

where e^w denotes the complex exponential function. Then:

1. The image of ϕ is the punctured unit disk

$$D^* = \{q \in \mathbb{C} : 0 < |q| < 1\},$$

and $\phi : \mathbb{H} \rightarrow D^*$ is a local biholomorphism whose fibres are precisely the orbits of the translation $z \mapsto z + h$.

2. There exists a unique holomorphic function $g : D^* \rightarrow \mathbb{C}$ such that

$$f(z) = g(e^{2\pi iz/h}) \quad \text{for all } z \in \mathbb{H}.$$

3. The function g admits a Laurent expansion

$$g(q) = \sum_{n=-\infty}^{\infty} a_n q^n$$

valid for all $q \in D^*$, which converges absolutely and uniformly on every annulus $0 < r_1 \leq |q| \leq r_2 < 1$. The coefficients a_n are given by the contour integral

$$a_n = \frac{1}{2\pi i} \oint_{|q|=\rho} \frac{g(q)}{q^{n+1}} dq$$

for any $\rho \in (0, 1)$.

4. Substituting $q = e^{2\pi iz/h}$, the Laurent expansion of g becomes the Fourier expansion of f (Definition 1.0.14)

$$f(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi inz/h},$$

converging absolutely and uniformly on every compact subset of $\mathbb{H}$. The coefficients a_n coincide with the Fourier coefficients defined by the contour integral formula

$$a_n = \frac{1}{h} \int_{z_0}^{z_0+h} f(z) e^{-2\pi inz/h} dz.$$

Proof. (♠ TODO: AI generated, verify) We prove each part in order.

1. For $z = x + iy \in \mathbb{H}$ with $y > 0$, we have

$$|\phi(z)| = |e^{2\pi i(x+iy)/h}| = e^{-2\pi y/h} < 1,$$

so $\phi(z) \in D^*$. Conversely, for any $q \in D^*$, write $q = re^{i\theta}$ with $0 < r < 1$ and $\theta \in \mathbb{R}$. Then $z = \frac{h}{2\pi}(\theta - i \log r)$ satisfies $\phi(z) = q$, and $\Im(z) = -\frac{h}{2\pi} \log r > 0$, so $z \in \mathbb{H}$. Hence $\phi(\mathbb{H}) = D^*$.

The derivative $\phi'(z) = \frac{2\pi i}{h} e^{2\pi iz/h}$ never vanishes on $\mathbb{H}$ because the exponential function never vanishes. By ?? and the inverse function theorem, ϕ is a local biholomorphism. Moreover, $\phi(z_1) = \phi(z_2)$ if and only if $e^{2\pi i(z_1 - z_2)/h} = 1$, which holds precisely when $z_1 - z_2 \in h\mathbb{Z}$. Thus the fibres of ϕ are the orbits of the translation $z \mapsto z + h$.

2. Since $f(z + h) = f(z)$, the function f is constant on each fibre of ϕ . Because ϕ is a local biholomorphism, the universal property of the quotient by a local biholomorphism implies that there exists a unique function $g : D^* \rightarrow \mathbb{C}$ such that $f = g \circ \phi$. To see that g is holomorphic, fix $q_0 \in D^*$ and choose $z_0 \in \phi^{-1}(q_0)$. Let U be a neighbourhood of z_0 on which ϕ restricts to a biholomorphism $\phi|_U : U \xrightarrow{\sim} V$. Then $g|_V = f \circ (\phi|_U)^{-1}$, which is a composition of holomorphic maps and hence holomorphic on V . Thus g is holomorphic on D^* .
3. The function g is holomorphic on the punctured disk $D^* = D(0, 1) \setminus \{0\}$. By ??, g admits a Laurent expansion

$$g(q) = \sum_{n=-\infty}^{\infty} a_n q^n$$

valid for all $q \in D^*$, converging absolutely and uniformly on every closed annulus $0 < r_1 \leq |q| \leq r_2 < 1$. The coefficients are given by the contour integral formula.

4. Substituting $q = e^{2\pi iz/h}$ yields the Fourier expansion of f . The convergence properties follow from the convergence of the Laurent expansion of g : for any compact subset $K \subset \mathbb{H}$, the image $\phi(K)$ is contained in a closed annulus $r_1 \leq |q| \leq r_2$ with $0 < r_1 < r_2 < 1$ (since $\Im(z)$ is bounded away from 0 and ∞ on K), and the Laurent series converges uniformly on that annulus.

It remains to verify the coefficient formula. Applying the Laurent coefficient formula to g and pulling back the contour integral via ϕ , we have for any $\rho \in (0, 1)$:

$$\begin{aligned} a_n &= \frac{1}{2\pi i} \oint_{|q|=\rho} \frac{g(q)}{q^{n+1}} dq \\ &= \frac{1}{2\pi i} \int_{z_0}^{z_0+h} \frac{g(e^{2\pi iz/h})}{e^{2\pi i(n+1)z/h}} \cdot \frac{2\pi i}{h} e^{2\pi iz/h} dz \\ &= \frac{1}{h} \int_{z_0}^{z_0+h} f(z) e^{-2\pi in z/h} dz, \end{aligned}$$

where the path is the image under ϕ^{-1} of the circle $|q| = \rho$, which is a horizontal segment of length h in $\mathbb{H}$. The independence of the choice of path and of z_0 follows from ??. Thus the coefficients a_n coincide with the Fourier coefficients defined in Definition 1.0.14.

□

Definition 1.0.16. (♠ TODO: define holomorphic function, upper half plane) (♠ TODO: define cusps of Γ) Let k be an integer and $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup (Definition 1.0.1). A *modular form of weight k with respect to Γ* is a function $f : \mathbb{H} \rightarrow \mathbb{C}$, where $\mathbb{H}$ is the upper half-plane, such that

- The function f is holomorphic on $\mathbb{H}$.
- The function f is a weakly modular form of weight k with respect to Γ (Definition 1.0.11). Equivalently, for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and all $z \in \mathbb{H}$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$

- f is holomorphic at the cusps of Γ (i.e., its Fourier expansion at all cusps of Γ is holomorphic). More precisely, $f|_k \alpha$ (Definition 1.0.10) is holomorphic at ∞ (Definition 1.0.13) for all $\alpha \in \Gamma$.

A *cusp form* is a modular form f whose Fourier expansion (Definition 1.0.14) at every cusp of Γ has constant term 0, i.e. $f|_k \alpha$ has a cusp at ∞ for every $\alpha \in \Gamma$.

Definition 1.0.17. (♠ TODO: define holomorphic function, upper half plane) (♠ TODO: define cusps of Γ) Let k be an integer and $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup (Definition 1.0.1). A *modular form of weight k with respect to Γ* is a function $f : \mathbb{H} \rightarrow \mathbb{C}$, where $\mathbb{H}$ is the upper half-plane, such that

- The function f is meromorphic on $\mathbb{H}$.
- The function f is a weakly modular form of weight k with respect to Γ (Definition 1.0.11). Equivalently, for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and all $z \in \mathbb{H}$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$

- f is meromorphic at the cusps of Γ (i.e., its Fourier expansion at all cusps of Γ is holomorphic). More precisely, $f|_k \alpha$ (Definition 1.0.10) is meromorphic at ∞ (Definition 1.0.13) for all $\alpha \in \Gamma$.

We define the L -function of a modular form by putting the Fourier coefficients of the modular form into a Dirichlet series.

Definition 1.0.18 (L-function of a modular form). Let $k \in \mathbb{Z}_{>0}$, and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a modular form (Definition 1.0.16) of weight k with respect to a congruence subgroup (Definition 1.0.1) $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$. Let

$$f(\tau) = \sum_{n=0}^{\infty} a_n e^{\frac{2\pi i n \tau}{h}}$$

be the Fourier expansion of f (Definition 1.0.13) where $h > 0$ is an appropriate integer.

The *L-function of the modular form f* is the complex function defined for $s \in \mathbb{C}$ with sufficiently large $\operatorname{Re}(s)$ by the Dirichlet series

$$L(f, s) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}.$$

(♠ TODO: define the completed L -function)

The construction of the L -function of a modular form is not arbitrary — a Fourier transform involving the modular form yields the completed L -function.

Theorem 1.0.19. Let $k \in \mathbb{Z}_{>0}$, and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a modular form (Definition 1.0.16) of weight k with respect to a congruence subgroup (Definition 1.0.1) $\Gamma \subseteq \operatorname{SL}_2(\mathbb{Z})$. Let

$$f(\tau) = \sum_{n=0}^{\infty} a_n e^{\frac{2\pi i n \tau}{h}}$$

be the Fourier expansion of f (Definition 1.0.13) where $h > 0$ is an appropriate integer.

The Mellin transform of $f(i\tau)$ equals the completed L -function $\Lambda(f, s)$.

Proof. Let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a meromorphic function periodic with period $h > 0$, so

$$f(z + h) = f(z).$$

This implies that f factors through the variable

$$q_h := e^{2\pi i z/h}.$$

Define

$$\Phi(t) := f(it), \quad t > 0.$$

Using the periodicity, f has a Fourier expansion in q_h :

$$f(z) = \sum_{n=n_0}^{\infty} a_n q_h^n = \sum_{n=n_0}^{\infty} a_n e^{2\pi i n z/h}.$$

Substituting $z = it$, we have

$$\Phi(t) = f(it) = \sum_{n=n_0}^{\infty} a_n e^{-2\pi n t/h}.$$

Consider the Mellin transform integral with complex parameter s :

$$I(s) := \int_0^{\infty} \Phi(t) t^s \frac{dt}{t}.$$

For $\operatorname{Re}(s)$ sufficiently large, interchange sum and integral (justified by absolute convergence):

$$I(s) = \sum_{n=n_0}^{\infty} a_n \int_0^{\infty} e^{-2\pi n t/h} t^s \frac{dt}{t}.$$

Evaluate the integral inside for each fixed n . Change variables by setting $u = \frac{2\pi nt}{h}$, so $t = \frac{hu}{2\pi n}$, and

$$dt = \frac{h}{2\pi n} du.$$

Hence,

$$\int_0^\infty e^{-2\pi nt/h} t^s \frac{dt}{t} = \int_0^\infty e^{-u} \left(\frac{hu}{2\pi n} \right)^s \frac{du}{u} = \left(\frac{h}{2\pi n} \right)^s \int_0^\infty e^{-u} u^s \frac{du}{u}.$$

Recognizing the Gamma function, we get

$$\int_0^\infty e^{-u} u^s \frac{du}{u} = \Gamma(s).$$

Therefore,

$$I(s) = \Gamma(s) \left(\frac{h}{2\pi} \right)^s \sum_{n=n_0}^\infty a_n n^{-s}.$$

By definition, the completed L -function $\Lambda(f, s)$ associated to f is proportional to this Mellin transform expression:

$$\Lambda(f, s) = \Gamma(s) \left(\frac{h}{2\pi} \right)^s L(f, s),$$

where

$$L(f, s) := \sum_{n=n_0}^\infty \frac{a_n}{n^s}.$$

This establishes the integral representation of $\Lambda(f, s)$ as the Mellin transform of $\Phi(t) = f(it)$, adapted to period h . $\square$

1.1. Hecke operators and Petersson inner product.

Definition 1.1.1. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup (Definition 1.0.1). The set of all modular forms of weight k with respect to Γ (Definition 1.0.16) forms a complex vector space, denoted by

$$M_k(\Gamma).$$

The subset of those modular forms that are also cusp forms (Definition 1.0.16) is a linear subspace, denoted by

$$S_k(\Gamma) \subseteq M_k(\Gamma).$$

The quotient vector space

$$\mathcal{E}_k(\Gamma) = M_k(\Gamma)/S_k(\Gamma)$$

is called the *space of Eisenstein series of weight k for Γ* . (Not every element of $\mathcal{E}_k(\Gamma)$ is represented by an Eisenstein series in the classical sense, but the name reflects that the image of the standard Eisenstein series spans this quotient when it exists.)

For each cusp s of Γ , there is a well-defined *constant term functional*

$$\text{ct}_s : S_k(\Gamma)^\perp \longrightarrow \mathbb{C},$$

where $S_k(\Gamma)^\perp \subseteq M_k(\Gamma)$ denotes the subspace of modular forms whose Fourier expansion at every cusp other than s has zero constant term. The dimension of $\mathcal{E}_k(\Gamma)$ is bounded above by the number of cusps of Γ .

Modular form (Definition 1.0.16) Cusp (Definition 1.2.3)

Definition 1.1.2. (♠ TODO: AI generated, verify) Let G be a group and let $\Gamma \leq G$ be a subgroup. For $g \in G$, the (Γ, Γ) -double coset of g in G , denoted by $\Gamma g \Gamma$, is defined as

$$\Gamma g \Gamma = \{\gamma_1 g \gamma_2 : \gamma_1, \gamma_2 \in \Gamma\}.$$

Equivalently, $\Gamma g \Gamma$ is the orbit of g under the action of $\Gamma \times \Gamma$ on G given by $(\gamma_1, \gamma_2) \cdot x = \gamma_1 x \gamma_2^{-1}$ for $\gamma_1, \gamma_2 \in \Gamma$ and $x \in G$.

The set of all (Γ, Γ) -double cosets in G is denoted by $\Gamma \backslash G / \Gamma$. When $\Gamma g \Gamma = \Gamma g' \Gamma$, we write $g' \in \Gamma g \Gamma$.

For any right coset representative α of Γ in the double coset $\Gamma g \Gamma$, we have a decomposition

$$\Gamma g \Gamma = \bigsqcup_i \Gamma \alpha_i$$

into finitely or countably many disjoint right cosets, depending on whether $[\Gamma : \Gamma \cap g^{-1} \Gamma g]$ is finite.

Definition 1.1.3 (Hecke operators on modular forms for $\text{SL}_2(\mathbb{Z})$). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}$ and let $\Gamma = \text{SL}_2(\mathbb{Z})$. For each positive integer n , the *Hecke operator of level n* (or simply the *n -th Hecke operator*) is the linear map

$$T_n : M_k(\Gamma) \longrightarrow M_k(\Gamma)$$

defined on a modular form $f \in M_k(\Gamma)$ (Definition 1.1.1) by

$$(T_n f)(z) = n^{k-1} \sum_{ad=n, d>0, 0 \leq b < d} d^{-k} f \Big|_k \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} (z),$$

(Definition 1.0.10)

(♠ TODO: state and reference a lemma justifying this equivalence) Equivalently, using the double coset (Definition 1.1.2) description, for each $n \geq 1$,

$$\Gamma \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix} \Gamma = \bigsqcup_{ad=n, d>0, 0 \leq b < d} \Gamma \begin{pmatrix} a & b \\ 0 & d \end{pmatrix},$$

and the Hecke operator is given by summing over right coset representatives:

$$(T_n f)(z) = n^{k-1} \sum_{\gamma \in \Gamma_\infty \backslash \Gamma \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix} \Gamma / \Gamma_\infty} (f|_k \gamma)(z),$$

where $\Gamma_\infty = \{\pm \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} : m \in \mathbb{Z}\}$ is the stabilizer of ∞ in Γ .

For a modular form $f \in M_k(\Gamma)$ (Definition 1.1.1) with Fourier expansion (Definition 1.0.14) $f(z) = \sum_{m=0}^{\infty} a_m q^m$, the Fourier coefficients of $T_n f$ are given by

$$T_n f = \sum_{m=0}^{\infty} b_m q^m, \quad \text{where } b_m = \sum_{d|\gcd(m,n)} d^{k-1} a_{mn/d^2}.$$

In particular, if f is a cusp form (Definition 1.0.16), then $T_n f$ is also a cusp form (Definition 1.0.16), so each T_n restricts to an endomorphism

$$T_n : S_k(\Gamma) \rightarrow S_k(\Gamma).$$

(Definition 1.1.1)

Definition 1.1.4. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup (Definition 1.0.1). The *Petersson inner product* on the space of cusp forms (Definition 1.0.16) $S_k(\Gamma)$ (Definition 1.1.1) is defined by

$$\langle f, g \rangle = \iint_{\Gamma \backslash \mathbb{H}} f(z) \overline{g(z)} y^k \frac{dx dy}{y^2},$$

(Definition 1.0.8) (♠ TODO: I'd like a reference associated to the double integral here. In what sense does this integral make sense? Is it supposed to be a Lebesgue integral? Precisely how should the notion of $dx dy$ be made sense as a measure on $\Gamma \backslash \mathbb{H}$ then?) where $z = x + iy$ and the integral is taken over a fundamental domain for Γ acting on the upper half-plane $\mathbb{H}$ via the modular action (Definition 1.0.8).

The measure $y^{-2} dx dy$ is the unique (up to scalar) $\mathrm{SL}_2(\mathbb{R})$ -invariant measure on $\mathbb{H}$ (Definition 1.0.4), so the integrand $f(z) \overline{g(z)} y^k y^{-2} dx dy$ is well-defined and independent of the choice of fundamental domain. (♠ TODO: define fundamental domain and reference it)

For cusp forms $f, g \in S_k(\Gamma)$, this integral converges absolutely, and $\langle \cdot, \cdot \rangle$ defines a positive-definite Hermitian inner product on $S_k(\Gamma)$. (♠ TODO: define the notion of Hermitian inner product and reference it — how is it related to hermitian form?)

Definition 1.1.5. (♠ TODO: AI generated, verify) (♠ TODO: define the notion of primitivity for dirichlet characters) Let N be a positive integer and let $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ be a group character, viewed as a primitive Dirichlet character modulo N .

A function $f : \mathbb{H} \rightarrow \mathbb{C}$ is called a *modular form of weight k with nebentypus χ for $\Gamma_1(N)$* if:

1. f is holomorphic on the upper half-plane (Definition 1.0.4) $\mathbb{H}$.
2. For every $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N)$, we have

$$f|_k \gamma = \chi(d) \cdot f,$$

where $f|_k \gamma$ is the weight- k slash operator (Definition 1.0.10).

3. For every $\sigma \in \mathrm{SL}_2(\mathbb{Z})$, the function $f|_k \sigma$ is holomorphic at ∞ (Definition 1.0.13).

The space of such modular forms is denoted by

$$M_k(\Gamma_1(N), \chi).$$

(♠ TODO: define the notion of fourier expansion at a cusp) A form $f \in M_k(\Gamma_1(N), \chi)$ is called a *cusp form with nebentypus χ* if its Fourier expansion at every cusp has zero constant term. The space of such forms is denoted by

$$S_k(\Gamma_1(N), \chi) \subseteq M_k(\Gamma_1(N), \chi).$$

When χ is the trivial character, $M_k(\Gamma_1(N), \chi)$ coincides with the space $M_k(\Gamma_0(N))$ of modular forms for $\Gamma_0(N)$ as defined by the general definition of modular forms for congruence subgroups (Definition 1.0.16). Cusp (Definition 1.2.3)

Theorem 1.1.6. (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}$ and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. For each positive integer n , the Hecke operator (Definition 1.1.3) T_n is a well-defined linear map

$$T_n : M_k(\Gamma) \longrightarrow M_k(\Gamma),$$

and it restricts to an endomorphism $T_n : S_k(\Gamma) \rightarrow S_k(\Gamma)$ on the subspace of cusp forms (Definition 1.0.16).

Proof. We first show that T_n is well defined as a map from $M_k(\Gamma)$ to itself. By definition, for $f \in M_k(\Gamma)$, we have

$$\begin{aligned} (T_n f)(z) &= n^{k-1} \sum_{ad=n, d>0, 0 \leq b < d} d^{-k} f \Big|_k \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} (z) \\ &= n^{k-1} \sum_{ad=n, d>0, 0 \leq b < d} d^{-k} \cdot d^{-k} f \left(\frac{az+b}{d} \right) \\ &= n^{k-1} \sum_{ad=n, d>0, 0 \leq b < d} d^{-2k} f \left(\frac{az+b}{d} \right) \end{aligned}$$

(Definition 1.0.10) The above is a finite sum, $z \mapsto \frac{az+b}{d}$ is holomorphic on $\mathbb{H}$, and f is holomorphic on $\mathbb{H}$, so the above is holomorphic on $\mathbb{H}$.

We need to show that $T_n f$ is a weakly modular form (Definition 1.0.11) of weight k with respect to Γ , i.e. that

$$((T_n f)|_k \gamma)(z) = (T_n f)(z)$$

for all $\gamma \in \Gamma$.

□

Proposition 1.1.7. (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}$ and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. The Hecke operators acting on the space of modular forms (Definition 1.1.1) $M_k(\Gamma)$ satisfy the following relations for all positive integers m, n :

1. *Multiplicativity for coprime indices:* If $(m, n) = 1$, then

$$T_m T_n = T_{mn}.$$

2. **Prime power recurrence:** For any prime p and any integer $r \geq 1$,

$$T_p^r T_p = T_p^{r+1} + p^{k-1} T_p^{r-1}.$$

3. **Commutativity:** For all $m, n \geq 1$, the operators commute:

$$T_m T_n = T_n T_m.$$

As a consequence of (1) and (2), the (♠ TODO: AI generated, reference needed: definition of Hecke algebra generated by Hecke operators) Hecke algebra generated by $\{T_n : n \geq 1\}$ is commutative, and every T_n can be expressed as a polynomial in the prime power operators $\{T_{p^r}\}$ using the relations above.

Proposition 1.1.8. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. For each prime p , the Hecke operator T_p is self-adjoint with respect to the Petersson inner product (Definition 1.1.4) on the space of cusp forms (Definition 1.1.1) $S_k(\Gamma)$: for all f, g in this space,

$$\langle T_p f, g \rangle = \langle f, T_p g \rangle.$$

As a consequence, the operators $\{T_p : p \text{ prime}\}$ are simultaneously diagonalizable (♠ TODO: AI generated, reference needed: theorem on simultaneous diagonalization of commuting self-adjoint operators on finite-dimensional inner product space) on $S_k(\Gamma)$ with respect to an orthonormal basis (♠ TODO: AI generated, reference needed: definition of orthonormal basis in an inner product space) of eigenforms.

Corollary 1.1.9. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. Let f in the space of cusp forms (Definition 1.1.1) $S_k(\Gamma)$ be a nonzero cusp form (Definition 1.0.16) that is a simultaneous eigenform for all Hecke operators, i.e., there exist complex numbers λ_n such that

$$T_n f = \lambda_n \cdot f \quad \text{for all } n \geq 1.$$

Normalize f so that its first Fourier coefficient is $a_1 = 1$. Then:

1. The eigenvalues satisfy $\lambda_n = a_n$ for all $n \geq 1$, where a_n are the Fourier coefficients of $f(z) = \sum_{n=0}^{\infty} a_n q^n$.
2. The eigenvalues (and Fourier coefficients) are multiplicative: if $(m, n) = 1$, then

$$a_{mn} = a_m a_n.$$

3. The eigenvalues satisfy the recurrence relation for each prime p and $r \geq 1$:

$$a_p^r \cdot a_p = a_{p^{r+1}} + p^{k-1} a_p^{r-1}.$$

4. The L-function of f (Definition 1.0.18) admits an Euler product expansion:

$$L(f, s) = \sum_{n=1}^{\infty} a_n n^{-s} = \prod_p (1 - a_p p^{-s} + p^{k-1-2s})^{-1},$$

which converges absolutely for $\mathrm{Re}(s)$ sufficiently large.

5. The Fourier coefficients a_n are algebraic integers for all $n \geq 1$.

Proposition 1.1.10. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. Let f, g in the space of cusp forms (Definition 1.1.1) $S_k(\Gamma)$ be nonzero simultaneous eigenforms for all Hecke operators with eigenvalues $\{\lambda_n(f)\}_{n \geq 1}$ and $\{\lambda_n(g)\}_{n \geq 1}$ respectively. If there exists some positive integer m such that

$$\lambda_m(f) \neq \lambda_m(g),$$

then f and g are orthogonal with respect to the Petersson inner product (Definition 1.1.4):

$$\langle f, g \rangle = 0.$$

In particular, if $\lambda_p(f) \neq \lambda_p(g)$ for some prime p , then $\langle f, g \rangle = 0$.

1.2. Modular curves.

Definition 1.2.1 (Classical affine modular curve associated to a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$). Let N be a positive integer and let H be a congruence subgroup (Definition 1.0.1) of $\mathrm{SL}_2(\mathbb{Z})$. The *classical affine modular curve associated to H* is the Riemann surface

$$Y_H = H \backslash \mathbb{H},$$

where $\mathbb{H}$ is the complex upper half-plane (Definition 1.0.4) and H acts via the modular action (Definition 1.0.8). Concretely, Y_H is the quotient of $\mathbb{H}$ by H equipped with the unique Riemann surface structure making the projection $\mathbb{H} \rightarrow Y_H$ holomorphic.

Definition 1.2.2 (Classical compactified modular curve associated to a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$). Let N be a positive integer and let H be a congruence subgroup (Definition 1.0.1) of $\mathrm{SL}_2(\mathbb{Z})$. The *classical compactified modular curve associated to H* is the Riemann surface

$$X_H = H \backslash (\mathbb{H} \cup \mathbb{P}^1(\mathbb{Q})),$$

obtained by adjoining to $Y_H = H \backslash \mathbb{H}$ the set of cusps (Definition 1.2.3). Each X_H is a compact Riemann surface, and the affine curve Y_H is the complement of finitely many points (the cusps) in X_H .

These compactified modular curves carry the structure of algebraic curves over $\mathbb{C}$ (and, by uniformization theory, over number fields), and the complex analytic isomorphisms above identify them with their associated Riemann surfaces.

Definition 1.2.3 (Cusps of classical modular curves). Let H be a congruence subgroup (Definition 1.0.1) of $\mathrm{SL}_2(\mathbb{Z})$. The *cusps of the classical modular curve associated to H* are the orbits of $\mathbb{P}^1(\mathbb{Q})$ under the modular action (Definition 1.0.8) of H , i.e. the elements of the quotient set

$$H \backslash \mathbb{P}^1(\mathbb{Q}).$$

This set of cusps is regarded as a subset of the extended upper half-plane $\hat{\mathbb{H}} = \mathbb{H} \cup \mathbb{P}^1(\mathbb{R})$. The cusps are finite in number and correspond to the points adjoined to the affine curve $Y_H = H \backslash \mathbb{H}$ to form the compactified modular curve X_H .

Definition 1.2.4. Let N be a positive integer. The *classical modular curve* $X_0(N)$ is the compact Riemann surface obtained as the quotient $X_0(N) := \Gamma_0(N) \backslash \overline{\mathbb{H}}$ where $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ is the upper half-plane (Definition 1.0.4) and $\Gamma_0(N) \subset \text{SL}_2(\mathbb{Z})$ (??) is the congruence subgroup.

The curve $X_0(N)$ classifies complex elliptic curves equipped with a cyclic subgroup of order N .

Definition 1.2.5. Let N be a positive integer. The *classical modular curve* $X(N)$ is the compact Riemann surface obtained as the quotient $X(N) := \Gamma(N) \backslash \overline{\mathbb{H}}$ where $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ is the upper half-plane (Definition 1.0.4) and $\Gamma(N) \subset \text{SL}_2(\mathbb{Z})$ is the principal congruence subgroup (Definition 1.0.2) of level N .

The curve $X(N)$ classifies complex elliptic curves $E/\mathbb{C}$ equipped with a full level N structure, i.e., an ordered basis of the N -torsion points $E[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. The open modular curve $Y(N) = \Gamma(N) \backslash \mathbb{H}$ is the non-compact Riemann surface before compactification.

Definition 1.2.6 (Compactified modular curve $X_0(N)$). Let $N \geq 1$ be an integer. The *affine modular curve* $Y_0(N)$ is the affine scheme over $\mathbb{Z}[1/N]$ that represents the moduli stack $\mathcal{M}_0(N)$ (see Moduli Stack $\mathcal{M}_0(N)$) for $N \geq 5$.

More precisely, $Y_0(N)$ is the fine moduli space representing the functor

$$Y_0(N) : (\text{Sch}/\mathbb{Z}[1/N])^{\text{op}} \longrightarrow \text{Sets}$$

sending an $\mathbb{Z}[1/N]$ -scheme S to the set of isomorphism classes of pairs (E, C) where

- E is an elliptic curve over S , and
- $C \subset E[S]$ is a cyclic subgroup scheme of order N (see cyclic subgroup scheme of order N).

For $N \geq 5$, every elliptic curve with a cyclic subgroup scheme of order N has no non-trivial automorphisms preserving the subgroup, so the moduli stack $\mathcal{M}_0(N)$ is represented by a fine moduli space. This representing object is the affine scheme $Y_0(N)$ over $\mathbb{Z}[1/N]$.

The *compactified modular curve* $X_0(N)$ is the proper smooth algebraic curve containing $Y_0(N)$ as a dense open subscheme. The complement $X_0(N) \setminus Y_0(N)$ consists of finitely many cusps (Definition 1.2.3), each corresponding to a Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ equipped with a cyclic subgroup scheme of order N in its generic fiber.

There is a natural *Atkin-Lehner involution* $w_N : X_0(N) \rightarrow X_0(N)$ sending (E, C) to $(E/C, E[N]/C)$, which exchanges the two cusps ∞ and 0 when N is prime.

Definition 1.2.7. Let N be a positive integer and let $H \subset \text{GL}_2(\widehat{\mathbb{Z}})$ be an open subgroup satisfying $\Gamma(N) \subseteq H$, where $\widehat{\mathbb{Z}} = \prod_{\ell} \mathbb{Z}_{\ell}$ is the profinite completion of $\mathbb{Z}$ and $\Gamma(N)$ is the principal congruence subgroup of level N viewed as a subgroup of $\text{GL}_2(\widehat{\mathbb{Z}})$ via the diagonal embedding. The *intermediate modular curve* X_H is the compact Riem surface obtained as the quotient $X_H := H \backslash \overline{\mathbb{H}}$ where $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ is the upper half-plane and the action of H on $\mathbb{H}$ is via the projection $H \rightarrow H \bmod N \subset \text{GL}_2(\mathbb{Z}/N\mathbb{Z})$ followed by the usual Möbius

action of $\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$ on $\Gamma(N)\backslash\mathbb{H}$. The curve X_H classifies elliptic curves equipped with an H -structure, i.e., a level structure whose stabiliser in $\mathrm{GL}_2(\widehat{\mathbb{Z}})$ is exactly H . The open modular curve $Y_H = H\backslash\mathbb{H}$ is the non-compact Riemann surface before compactification. Special cases include $X_{\Gamma(N)} = X(N)$, $X_{\Gamma_0(N)} = X_0(N)$, and $X_{\Gamma_1(N)} = X_1(N)$.

Definition 1.2.8 (Compactified modular curve $X(N)$). Let $N \geq 1$ be an integer. The *compactified modular curve* $X(N)$ is the proper smooth algebraic curve over $\mathbb{Z}[1/N, \zeta_N]$ that compactifies the affine fine moduli space $Y(N)$ representing the moduli stack $\mathcal{M}(N)$ (see Moduli Stack $\mathcal{M}(N)$).

More precisely, for $N \geq 3$, the affine modular curve $Y(N)$ is the fine moduli space representing the functor

$$Y(N) : (\mathrm{Sch}/\mathbb{Z}[1/N, \zeta_N])^{\mathrm{op}} \longrightarrow \mathrm{Sets}$$

sending an $\mathbb{Z}[1/N, \zeta_N]$ -scheme S to the set of isomorphism classes of pairs (E, α) where

- E is an elliptic curve over S , and
- $\alpha : (\mathbb{Z}/N\mathbb{Z})_S^{\oplus 2} \xrightarrow{\sim} E[N]$ is a full level N structure (see full level N structure).

The compactified curve $X(N)$ is obtained by adjoining finitely many cusps (Definition 1.2.3) to $Y(N)$. Each cusp corresponds to a Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ over a complete discrete valuation field, equipped with a full level N structure on its generic fiber.

The Galois group $\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})/\{\pm I\}$ acts on $X(N)$ by change of basis of the full level structure. The quotient of $X(N)$ by the image of $\Gamma_1(N) \subset \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ is the compactified modular curve $X_1(N)$, and the quotient by the image of $\Gamma_0(N)$ is the compactified modular curve $X_0(N)$.

Definition 1.2.9 (Compactified modular curve $X_s(N)$). Let $N \geq 1$ be an integer and let $\Gamma \subset \mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$ be a subgroup that is not the image of any congruence subgroup (Definition 1.0.1) of $\mathrm{SL}_2(\mathbb{Z})$. Such a subgroup is called a *non-congruence subgroup*. The *non-classical modular curve* $X_s(N)$ associated to Γ is the compact Riemann surface (or complete algebraic curve) defined as the quotient

$$X_s(N) = \Gamma \backslash \mathbb{H}^*$$

where $\mathbb{H}^* = \mathbb{H} \cup \mathbb{P}^1(\mathbb{Q})$ is the extended upper half-plane and Γ acts via the Möbius action.

More precisely, $X_s(N)$ is obtained by taking the quotient of the compactified modular curve $X(N)$ (Definition 1.2.8) by the action of Γ on the full level N structures. The resulting curve is a finite cover of $X(1)$, but unlike the classical modular curves $X_0(N)$, $X_1(N)$, and $X(N)$, it does not parameterize elliptic curves with a moduli-theoretically natural level structure arising from a congruence subgroup.

The curve $X_s(N)$ was studied by Serre as an example of a modular curve that does not arise from the classical congruence subgroup theory.

1.3. Eisenstein series.

Definition 1.3.1 (Series indexed by a countable set). (♠ TODO: AI generated, verify) Let I be a countable set and $(a_i)_{i \in I}$ a family of real numbers.

The series $\sum_{i \in I} a_i$ is said to *converge absolutely* if the supremum over all finite subsets is finite:

$$\sum_{i \in I} |a_i| := \sup_{\substack{F \subseteq I \\ |F| < \infty}} \sum_{i \in F} |a_i| < \infty.$$

In this case, Lemma 1.3.2 guarantees that for every bijection $\varphi : \mathbb{N} \rightarrow I$, the ordinary series $\sum_{n=1}^{\infty} a_{\varphi(n)}$ converges (absolutely) and to the same sum independent of the enumeration; this common sum is denoted

$$\sum_{i \in I} a_i.$$

When the terms are nonnegative, the sum reduces to the supremum above, and the series converges precisely when this supremum is finite.

Lemma 1.3.2 (Absolute convergence over a countable index set is unconditional). (♠ **TODO: AI generated, verify**) Let V be a Banach space over $\mathbb{R}$ or $\mathbb{C}$, with norm $\|\cdot\|$. Let I be a countable set and $(v_i)_{i \in I}$ a family in V such that

$$\sum_{i \in I} \|v_i\| := \sup_{\substack{F \subseteq I \\ |F| < \infty}} \sum_{i \in F} \|v_i\| < \infty.$$

Then the following hold:

1. Absolute convergence for any enumeration: For every bijection $\varphi : \mathbb{N} \rightarrow I$, the ordinary series $\sum_{n=1}^{\infty} v_{\varphi(n)}$ converges absolutely in V ; consequently it converges in V by ??.
2. Uniqueness of the sum: For any two bijections $\varphi, \psi : \mathbb{N} \rightarrow I$,

$$\sum_{n=1}^{\infty} v_{\varphi(n)} = \sum_{n=1}^{\infty} v_{\psi(n)}.$$

In particular, for an absolutely convergent family $(v_i)_{i \in I}$, the common sum is well-defined and is denoted $\sum_{i \in I} v_i$.

Proof. (♠ **TODO: AI generated, verify**) Let $S = \sum_{i \in I} \|v_i\| < \infty$ be the supremum.

1. Let $\varphi : \mathbb{N} \rightarrow I$ be any bijection. For each N , the set $\{\varphi(1), \dots, \varphi(N)\}$ is a finite subset of I , so

$$\sum_{n=1}^N \|v_{\varphi(n)}\| \leq S.$$

Hence the sequence of partial sums of $\sum_{n=1}^{\infty} \|v_{\varphi(n)}\|$ is bounded above by S , so the series of norms converges to some value $\leq S$. Thus $\sum_{n=1}^{\infty} v_{\varphi(n)}$ converges absolutely, and by ?? it converges in V .

2. Let $\varphi, \psi : \mathbb{N} \rightarrow I$ be two bijections. Denote

$$S_{\varphi} = \sum_{n=1}^{\infty} \|v_{\varphi(n)}\|, \quad S_{\psi} = \sum_{n=1}^{\infty} \|v_{\psi(n)}\|,$$

which exist by part (1). Let $\varepsilon > 0$ be given. Since $S = \sum_{i \in I} \|v_i\|$ is finite, there exists a finite subset $F \subseteq I$ such that

$$\sum_{i \in I \setminus F} \|v_i\| < \varepsilon.$$

Choose N large enough that $F \subseteq \{\varphi(1), \dots, \varphi(N)\}$ and $F \subseteq \{\psi(1), \dots, \psi(N)\}$, which is possible because φ and ψ are bijections. For this N , the symmetric differences

$$\varphi(\{1, \dots, N\}) \setminus \psi(\{1, \dots, N\}) \quad \text{and} \quad \psi(\{1, \dots, N\}) \setminus \varphi(\{1, \dots, N\})$$

are both contained in $I \setminus F$, because every element of F belongs to both sets. Hence

$$\begin{aligned} \left\| \sum_{n=1}^N v_{\varphi(n)} - \sum_{n=1}^N v_{\psi(n)} \right\| &\leq \sum_{i \in \varphi(\{1, \dots, N\}) \setminus \psi(\{1, \dots, N\})} \|v_i\| + \sum_{i \in \psi(\{1, \dots, N\}) \setminus \varphi(\{1, \dots, N\})} \|v_i\| \\ &\leq 2 \sum_{i \in I \setminus F} \|v_i\| < 2\varepsilon. \end{aligned}$$

By part (1), $\sum_{n=1}^N v_{\varphi(n)} \rightarrow S_\varphi$ and $\sum_{n=1}^N v_{\psi(n)} \rightarrow S_\psi$ as $N \rightarrow \infty$. Choose N large enough that additionally

$$\left\| \sum_{n=1}^N v_{\varphi(n)} - S_\varphi \right\| < \varepsilon \quad \text{and} \quad \left\| \sum_{n=1}^N v_{\psi(n)} - S_\psi \right\| < \varepsilon.$$

Then by the triangle inequality,

$$\begin{aligned} \|S_\varphi - S_\psi\| &\leq \|S_\varphi - \sum_{n=1}^N v_{\varphi(n)}\| + \left\| \sum_{n=1}^N v_{\varphi(n)} - \sum_{n=1}^N v_{\psi(n)} \right\| + \left\| \sum_{n=1}^N v_{\psi(n)} - S_\psi \right\| \\ &< \varepsilon + 2\varepsilon + \varepsilon = 4\varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, $\|S_\varphi - S_\psi\| = 0$, so $S_\varphi = S_\psi$.

□

Theorem 1.3.3 (Comparison of double series to double integrals of monotone functions).
(♠ TODO: AI generated, verify) Let $f : [0, \infty)^2 \rightarrow [0, \infty)$ be a function that is nonincreasing in each variable separately, i.e. for all $x_1 \leq x_2$ and $y_1 \leq y_2$,

$$f(x_1, y_1) \geq f(x_2, y_1) \quad \text{and} \quad f(x_1, y_1) \geq f(x_1, y_2).$$

1. For any integers $m_0, n_0 \geq 0$,

$$\sum_{m=m_0+1}^{\infty} \sum_{n=n_0+1}^{\infty} f(m, n) \leq \iint_{[m_0, \infty) \times [n_0, \infty)} f(x, y) \, dx \, dy \leq \sum_{m=m_0}^{\infty} \sum_{n=n_0}^{\infty} f(m, n)$$

, where the integrals are taken with respect to Lebesgue measure on $\mathbb{R}^2$.

2. In particular, the double series $\sum_{(m,n) \neq (0,0)} f(m, n)$ converges if and only if the double integral $\iint_{[0, \infty)^2} f(x, y) \, dx \, dy$ is finite.

Proof. (♠ TODO: AI generated, verify) The monotonicity of f implies that for any $(m, n) \in \mathbb{Z}_{\geq 0}^2$,

$$f(m+1, n+1) \leq f(x, y) \leq f(m, n) \quad \text{for all } (x, y) \in [m, m+1] \times [n, n+1].$$

Summing these inequalities over $m \geq m_0$, $n \geq n_0$ and using the fact that the Lebesgue measure of the unit square $[m, m+1] \times [n, n+1]$ is 1, we obtain

$$\begin{aligned} \sum_{m=m_0+1}^{\infty} \sum_{n=n_0+1}^{\infty} f(m, n) &\leq \sum_{m=m_0}^{\infty} \sum_{n=n_0}^{\infty} \iint_{[m, m+1] \times [n, n+1]} f(x, y) dx dy \\ &\leq \sum_{m=m_0}^{\infty} \sum_{n=n_0}^{\infty} f(m, n). \end{aligned}$$

The middle double sum equals $\iint_{[m_0, \infty) \times [n_0, \infty)} f(x, y) dx dy$ by ?? (the terms are nonnegative, so the iterated integral equals the double integral). This proves (1).

Statement (2) follows from (1) by taking $m_0 = n_0 = 0$ and noting that the finiteness of the double series is equivalent to the finiteness of $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n)$ (since the boundary terms $m = 0$ or $n = 0$ contribute at most a finite number of additional terms). $\square$

Definition 1.3.4 (Polar coordinates on $\mathbb{R}^2$). (♠ TODO: AI generated, verify) The *polar coordinate map* is the function $\Phi : [0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}^2$ defined by

$$\Phi(r, \theta) = (r \cos \theta, r \sin \theta).$$

For a point $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, the standard restrictions are $r = \sqrt{x^2 + y^2} > 0$ (the *radial coordinate*) and θ (the *angular coordinate*) is any real number satisfying $x = r \cos \theta$ and $y = r \sin \theta$.

The restriction of Φ to $(0, \infty) \times (-\pi, \pi]$ is a bijection onto $\mathbb{R}^2 \setminus \{(0, 0)\}$. The *standard domain for polar coordinates* is the open set

$$U = (0, \infty) \times (-\pi, \pi) \subseteq \mathbb{R}^2,$$

on which Φ is a C^∞ diffeomorphism onto its image $\Phi(U) = \mathbb{R}^2 \setminus \{(x, 0) : x \leq 0\}$.

The *Jacobian determinant of the polar coordinate transformation* is

$$\det D\Phi(r, \theta) = r.$$

Theorem 1.3.5 (Change of variables for Lebesgue integrals on $\mathbb{R}^n$). (♠ TODO: AI generated, verify) Let $U, V \subseteq \mathbb{R}^n$ be open subsets and let $\Phi : U \rightarrow V$ be a C^1 diffeomorphism. For any Lebesgue measurable function $f : V \rightarrow [0, \infty]$, we have

$$\int_V f(y) d\lambda^n(y) = \int_U f(\Phi(x)) |\det D\Phi(x)| d\lambda^n(x),$$

where $D\Phi(x)$ denotes the *Jacobian matrix* of Φ at x and $|\det D\Phi(x)|$ is its absolute determinant.

If $f : V \rightarrow \overline{\mathbb{R}}$ is Lebesgue integrable, then the same equality holds, with both sides finite.

Definition 1.3.6 (Lebesgue integral over a measurable subset of $\mathbb{R}^n$). (♠ TODO: AI generated, verify) Let $E \subseteq \mathbb{R}^n$ be a Lebesgue measurable subset and let $f : E \rightarrow \overline{\mathbb{R}}$ be a measurable function.

The *Lebesgue integral of f over E* is defined by extending f by zero outside E :

$$\int_E f d\lambda^n := \int_{\mathbb{R}^n} \tilde{f} d\lambda^n,$$

where $\tilde{f} : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is defined by $\tilde{f}(x) = f(x)$ for $x \in E$ and $\tilde{f}(x) = 0$ for $x \notin E$.

The integral is said to *converge* (or f is *integrable over E*) when $\int_E |f| d\lambda^n < \infty$.

In particular, for a nonnegative measurable function $f : E \rightarrow [0, \infty]$, the integral $\int_E f d\lambda^n$ is always defined as an element of $[0, \infty]$ (possibly infinite).

Definition 1.3.7 (Eisenstein series of weight k for a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$). (♠ TODO: AI generated, verify) Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ (Definition 1.0.1), let $\Gamma_\infty = \Gamma \cap \left\{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\}$ be the stabilizer of ∞ in Γ (Definition 1.0.9), and let $k \geq 3$ be an integer. Let $\mathbb{H}$ denote the upper half-plane (Definition 1.0.4).

The *Eisenstein series of weight k for Γ* is the function $E_k^\Gamma : \mathbb{H} \rightarrow \mathbb{C}$ defined by

$$E_k^\Gamma(z) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} j(\gamma, z)^{-k}.$$

Here $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $j(\gamma, z) = cz + d$ is the automorphy factor (Definition 1.0.10). The sum is taken over a set of representatives of the right cosets $\Gamma_\infty \backslash \Gamma$.

For $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, the coset space $\Gamma_\infty \backslash \mathrm{SL}_2(\mathbb{Z})$ is identified with the set of coprime pairs $(c, d) \in \mathbb{Z}^2$ up to sign, and the Eisenstein series reduces to the lattice sum

$$E_k^{\mathrm{SL}_2(\mathbb{Z})}(z) = \frac{1}{2} \sum_{\substack{(c,d) \in \mathbb{Z}^2 \\ (c,d)=1}} \frac{1}{(cz + d)^k},$$

where the factor $\frac{1}{2}$ accounts for the fact that (c, d) and $(-c, -d)$ represent the same coset. The *unnormalized Eisenstein series* is

$$G_k(z) = \sum_{\substack{(m,n) \in \mathbb{Z}^2 \\ (m,n) \neq (0,0)}} \frac{1}{(mz + n)^k}.$$

We have $G_k(z) = 2\zeta(k) E_k^{\mathrm{SL}_2(\mathbb{Z})}(z)$ where $\zeta(k)$ is the Riemann zeta function evaluated at k .

Theorem 1.3.8 (Eisenstein series for $\mathrm{SL}_2(\mathbb{Z})$ is a modular form of weight k). (♠ TODO: AI generated, verify) Let $k \geq 3$ be an integer. The unnormalized Eisenstein series $G_k(z)$

(Definition 1.3.7) for $\mathrm{SL}_2(\mathbb{Z})$ defines a modular form (Definition 1.0.16) of weight k for $\mathrm{SL}_2(\mathbb{Z})$. Its Fourier expansion (Definition 1.0.14) at the cusp (Definition 1.2.3) ∞ is given by

$$G_k(z) = 2\zeta(k) + 2 \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) e^{2\pi i n z},$$

(♠ TODO: create the definition of σ_k as its own file and reference it) where $\zeta(s)$ is the Riemann zeta function and $\sigma_{k-1}(n) = \sum_{d|n} d^{k-1}$ is the *sum of $(k-1)$ -st powers of divisors function*.

Consequently, the normalized Eisenstein series $E_k(z)$ (Definition 1.3.7) has the Fourier expansion

$$E_k(z) = 1 + \frac{(-2\pi i)^k}{(k-1)! \zeta(k)} \sum_{n=1}^{\infty} \sigma_{k-1}(n) e^{2\pi i n z}.$$

Proof. (♠ TODO: AI generated, verify) We verify each of the defining properties of a modular form.

1. We show that $G_k(z)$ converges absolutely and uniformly on compact subsets of $\mathbb{H}$. Let $K \subset \mathbb{H}$ be compact. For each $z = x + iy \in \mathbb{H}$,

$$|mz + n|^2 = |(mx + n) + imy|^2 = (mx + n)^2 + (my)^2 = m^2(x^2 + y^2) + 2mnx + n^2.$$

This is a positive definite quadratic form in (m, n) with matrix

$$\begin{pmatrix} |z|^2 & x \\ x & 1 \end{pmatrix}.$$

Its minimum eigenvalue

$$\lambda_{\min}(z) = \frac{|z|^2 + 1 - \sqrt{(|z|^2 - 1)^2 + 4x^2}}{2}$$

is positive because $y = \Im(z) > 0$ (the discriminant satisfies $(|z|^2 - 1)^2 + 4x^2 = (|z|^2 + 1)^2 - 4y^2 < (|z|^2 + 1)^2$, so $\lambda_{\min}(z) > 0$). The function $\lambda_{\min} : \mathbb{H} \rightarrow \mathbb{R}_{>0}$ is continuous, and K is compact, so

$$c := \min_{z \in K} \lambda_{\min}(z) > 0.$$

By the spectral theorem, for every $z \in K$ and every $(m, n) \in \mathbb{R}^2$,

$$|mz + n|^2 \geq \lambda_{\min}(z) \cdot (m^2 + n^2) \geq c(m^2 + n^2).$$

Consequently,

$$\frac{1}{|mz + n|^k} \leq \frac{1}{c^{k/2}} \cdot \frac{1}{(m^2 + n^2)^{k/2}}.$$

By Definition 1.3.1, the double series $\sum_{(m,n) \neq (0,0)} (m^2 + n^2)^{-k/2}$ is defined by the supremum of its finite partial sums. The function $(x, y) \mapsto (x^2 + y^2)^{-k/2}$ is nonincreasing in each variable separately on $[0, \infty)^2$. By Theorem 1.3.3 with $m_0 = n_0 = 1$, we have

$$\sum_{m=2}^{\infty} \sum_{n=2}^{\infty} (m^2 + n^2)^{-k/2} \leq \iint_{[1, \infty)^2} (x^2 + y^2)^{-k/2} dx dy.$$

(Definition 1.3.6) The remaining terms of the double series (those with $m = 0$, $m = 1$, $n = 0$, or $n = 1$) are bounded by the convergent single series $\sum_{m=1}^{\infty} m^{-k} = \zeta(k) < \infty$, and similarly for n .

To evaluate the double integral, apply Theorem 1.3.5 to the polar coordinate map (Definition 1.3.4) $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$. On the region $U = \{(r, \theta) \in (0, \infty) \times (0, \pi/2) : r \cos \theta \geq 1, r \sin \theta \geq 1\}$, the map Φ is a C^1 diffeomorphism onto $[1, \infty)^2$, and $|\det D\Phi(r, \theta)| = r$. Hence

$$\begin{aligned} \iint_{[1, \infty)^2} (x^2 + y^2)^{-k/2} dx dy &= \iint_U r^{-k} \cdot r dr d\theta \\ &\leq \int_0^{\pi/2} \int_1^{\infty} r^{1-k} dr d\theta = \frac{\pi}{2} \int_1^{\infty} r^{1-k} dr, \end{aligned}$$

which is finite for $k > 2$ because $\int_1^{\infty} r^{1-k} dr = 1/(k-2)$. Thus the double series converges. By the Weierstrass M -test, the series defining $G_k(z)$ converges absolutely and uniformly on K .

2. We show that $G_k(z)$ is Holomorphic on $\mathbb{H}$. Each term $(mz + n)^{-k}$ is a holomorphic function of z on $\mathbb{H}$ (it is a rational function with a simple pole at $z = -n/m$ for $m \neq 0$, which lies on the real axis, not in $\mathbb{H}$). The series converges uniformly on compact subsets of $\mathbb{H}$ by the previous step. By Theorem A.0.1, the limit function $G_k(z)$ is holomorphic on $\mathbb{H}$.
3. We verify that G_k is weakly modular (Definition 1.0.11). For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$, we compute:

$$\begin{aligned} G_k(\gamma z) &= \sum_{(m,n) \neq (0,0)} \frac{1}{(m \cdot \frac{az+b}{cz+d} + n)^k} \\ &= \sum_{(m,n) \neq (0,0)} \frac{(cz+d)^k}{(m(az+b) + n(cz+d))^k} \\ &= (cz+d)^k \sum_{(m,n) \neq (0,0)} \frac{1}{((ma+nc)z + (mb+nd))^k}. \end{aligned}$$

Since γ is an invertible 2×2 matrix of integers, the map $(m, n) \mapsto (m, n)\gamma = (ma + nc, mb + nd)$ is a permutation of $\mathbb{Z}^2 \setminus \{(0, 0)\}$. Hence the sum over (m, n) is identical to the sum over $(m', n') = (m, n)\gamma$, and we obtain

$$G_k(\gamma z) = (cz+d)^k \sum_{(m',n') \neq (0,0)} \frac{1}{(m'z + n')^k} = (cz+d)^k G_k(z).$$

Thus G_k satisfies the modular transformation law of weight k .

4. We show that $G_k(z)$ is holomorphic at ∞ (Definition 1.0.13) and derive its Fourier series. By Lemma 1.0.12, $G_k(z+1) = G_k(z)$ because $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Indeed,

$$G_k(z+1) = \sum_{(m,n) \neq (0,0)} \frac{1}{(m(z+1) + n)^k} = \sum_{(m,n) \neq (0,0)} \frac{1}{(mz + (m+n))^k} = G_k(z)$$

by the change of variables $(m, n) \mapsto (m, n - m)$.

By Theorem 1.0.15, G_k has a Fourier expansion

$$G_k(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n z}.$$

To compute the coefficients, we split the sum defining G_k into the $m = 0$ and $m \neq 0$ parts. Absolute convergence of the double series (established above) implies, by Lemma 1.3.2, that the sum is independent of the enumeration of $\mathbb{Z}^2 \setminus \{(0, 0)\}$. Grouping the terms as $m = 0$ then $m > 0$ then $m < 0$ specifies one such enumeration, so

$$G_k(z) = \sum_{n \neq 0} \frac{1}{n^k} + \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{(mz + n)^k} + \sum_{m=-\infty}^{-1} \sum_{n=-\infty}^{\infty} \frac{1}{(mz + n)^k}.$$

The first sum is $2 \sum_{n=1}^{\infty} n^{-k} = 2\zeta(k)$ (for k even, this is well-known; for odd k the sum still converges absolutely).

For the second sum, we apply Theorem A.0.2. For $m \geq 1$,

$$\sum_{n=-\infty}^{\infty} \frac{1}{(mz + n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m z},$$

and similarly for $m \leq -1$ with z replaced by $-|m|z$. Summing over all $m \neq 0$ collects the coefficients into divisor sums:

$$\begin{aligned} G_k(z) &= 2\zeta(k) + \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m z} + \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=-\infty}^{-1} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m z} \\ &= 2\zeta(k) + \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \left(\sum_{d|n} d^{k-1} \right) e^{2\pi i n z} \quad (\text{setting } n = \ell m) \\ &= 2\zeta(k) + 2 \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) e^{2\pi i n z}. \end{aligned}$$

(The factor of 2 arises because the $m = -\ell$ terms contribute the same sum.) This is the claimed Fourier expansion. In particular, there are no negative powers of $e^{2\pi i z}$, so G_k is holomorphic at ∞ by Definition 1.0.13.

Having verified all four properties – absolute convergence and uniform convergence on compacta yielding holomorphicity on $\mathbb{H}$, the modular transformation law, periodicity, and holomorphicity at ∞ with the explicit Fourier expansion – we conclude that G_k is a modular form of weight k for $\text{SL}_2(\mathbb{Z})$. $\square$

2. AUTOMORPHIC FORMS

Definition 2.0.1 (Adelic Quotient). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles (Definition B.0.4) of F . Let G be a linear algebraic group (Definition B.0.7) defined over F . The *adelic quotient of G over F* is the quotient space

$$G(F) \backslash G(\mathbb{A}_F).$$

(Definition B.0.5) where $G(F)$ is given the discrete topology. (♠ TODO: What topology does $G(F)$ have?)

Definition 2.0.2 (Automorphy). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F .

Let $f : G(\mathbb{A}_F) \rightarrow \mathbb{C}$ be a function. We say f satisfies *automorphy* if it is left $G(F)$ -invariant, i.e.,

$$f(\gamma g) = f(g) \quad \text{for all } \gamma \in G(F), g \in G(\mathbb{A}_F).$$

Equivalently, f is a function which descends to a well defined function $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ on the adelic quotient (Definition 2.0.1).

Definition 2.0.3 (Right-regular action of $G(\mathbb{A}_F)$). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles (Definition B.0.4) of F . Let G be a linear algebraic group (Definition B.0.7) defined over F . For $g \in G(\mathbb{A}_F)$ (Definition B.0.5) and a function $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$, the *right-regular action/representation of $G(\mathbb{A}_F)$* is the representation

$$R : G(\mathbb{A}_F) \rightarrow \text{Aut}(\text{Fun}(G(F) \backslash G(\mathbb{A}_F), \mathbb{C}))$$

on the space of complex valued functions on $G(F) \backslash G(\mathbb{A}_F)$ given by

$$(R(g)f)(x) = f(xg)$$

for $g \in G(\mathbb{A}_F)$, $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$, and $x \in G(F) \backslash G(\mathbb{A}_F)$.

Definition 2.0.4 (Smooth function). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F .

Let $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ be a function. We say f is *smooth* if it is (♠ TODO: define $G(\mathbb{A}_F)$, $G(\mathbb{A}_F^{\text{fin}})$, $G(F_\infty)$)

1. locally constant in the non-archimedean directions, i.e. for every point $g = (g_v)_v \in G(\mathbb{A}_F)$, there exists an open compact subgroup $U = \prod_v U_v \subset G(\mathbb{A}_F^{\text{fin}})$ (Definition B.0.5) such that for all $u \in U$, we have

$$f(gu) = f(g),$$

and

2. C^∞ in each archimedean direction, i.e. for every point $g = (g_v)_v \in G(\mathbb{A}_F)$, the map

$$G(F_\infty) \rightarrow \mathbb{C}, (h_v)_{v|\infty} \mapsto f(g \cdot (h_v)_{v|\infty})$$

(Definition B.0.5) is C^∞ as a smooth manifold; note that $G(F_\infty) = \prod_{v|\infty} G(F_v)$ is a product of real and complex Lie groups. (♠ TODO: define real and complex lie groups)

Definition 2.0.5 (K -finiteness). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F .

Let $K \subseteq G(\mathbb{A}_F)$ be a maximal compact subgroup. A smooth function (Definition 2.0.4) $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ is called ***K-finite*** if the vector space

$$\langle R(k)f : k \in K \rangle$$

(Definition 2.0.3) spanned by the K -translates of f under the right-regular action (Definition 2.0.3) is finite-dimensional.

Definition 2.0.6 (Moderate growth). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F . Let $\|\cdot\|$ be a fixed height function (or norm-like function) on $G(\mathbb{A}_F)$ compatible with F and G (e.g. the height function of Definition B.0.6).

A smooth function (Definition 2.0.4) $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ is said to have ***moderate growth*** if there exist constants $C > 0$ and $N \geq 0$ such that

$$|f(g)| \leq C \cdot \|g\|^N \quad \text{for all } g \in G(\mathbb{A}_F),$$

Definition 2.0.7 (Automorphic form). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F .

A function $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ is called an ***automorphic form*** if it satisfies all of the following conditions:

1. f is smooth (Definition 2.0.4),
2. f is K -finite (Definition 2.0.5) for some maximal compact subgroup $K \subseteq G(\mathbb{A}_F)$,
3. f has moderate growth (Definition 2.0.6),

(♠ TODO: read the following AI generated statements and make them precise) (♠ TODO: Formulate how modular forms are automorphic forms)

Theorem 2.0.8 (Basic Properties of Automorphic Forms). Let F be a global field, $\mathbb{A}_F$ its ring of adeles, and G a linear algebraic group defined over F . Let $K \subseteq G(\mathbb{A}_F)$ be a maximal compact subgroup. Suppose

$$f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$$

is an automorphic form, i.e., f is smooth, K -finite, and of moderate growth. Then:

1. f is left-invariant under $G(F)$ (automorphy condition).
2. f generates a $(\mathfrak{g}, K)$ -module under the right regular representation of $G(\mathbb{A}_F)$, where $\mathfrak{g}$ is the Lie algebra of $G(F_\infty)$.
3. The space of automorphic forms is preserved by Hecke operators arising from elements in the spherical Hecke algebra associated with $G(\mathbb{A}_F^\infty)$.

Proposition 2.0.9 (Equivalent Characterizations of Automorphic Forms). Under standard assumptions on G , the following conditions on a function $f : G(F) \backslash G(\mathbb{A}_F) \rightarrow \mathbb{C}$ are equivalent:

1. f is smooth, K -finite, of moderate growth, and left $G(F)$ -invariant.
2. f corresponds to a vector in a smooth admissible representation of $G(\mathbb{A}_F)$ realized as a subrepresentation of the space of L^2 automorphic forms.

Corollary 2.0.10 (Automorphic Forms as Sections). Automorphic forms can be identified with smooth sections of certain automorphic vector bundles over the quotient space $G(F)\backslash G(\mathbb{A}_F)/K$ satisfying growth conditions.

Remark 2.0.11 (Intuition). Automorphic forms generalize classical modular forms by encoding arithmetic and representation-theoretic data through functions on adelic quotients. The conditions of smoothness, K -finiteness, and moderate growth control analytic and algebraic properties needed for harmonic analysis and spectral theory on these quotients.

APPENDIX A. MISCELLANEOUS DEFINITIONS

Definition A.0.1 (Topological groups). A *topological group* is a group $(G, \cdot)$ together with a topology $\mathcal{T}$ on G such that the maps

$$\begin{aligned}\mu : G \times G &\rightarrow G, & (g, h) &\mapsto g \cdot h, \\ \iota : G &\rightarrow G, & g &\mapsto g^{-1},\end{aligned}$$

are continuous with respect to the product topology on $G \times G$ and the topology $\mathcal{T}$ on G .

A *abelian topological group* or a *commutative topological group* is a topological group that is abelian as a group.

Definition A.0.2. Let $n \in \mathbb{Z}_{>0}$. For $a, b \in \mathbb{Z}$, we say that a and b are *congruent modulo n* if $n \mid (a - b)$. This is denoted by

$$a \equiv b \pmod{n}.$$

If $n \nmid (a - b)$, we write $a \not\equiv b \pmod{n}$.

Definition A.0.3. (♠ TODO: AI generated, verify) Let (A, σ) be a ring with involution.

Two matrices $H, K \in M_n(A)$ are called **-congruent* (or *conjugate-transpose congruent*) if there exists an invertible matrix $P \in \text{GL}_n(A)$ such that

$$K = P^* H P,$$

where P^* denotes the conjugate transpose of P .

If H is an ε -Hermitian matrix and K is *-congruent to H , then K is also ε -Hermitian.

Let M be a free (right) A -module of rank n with an ε -Hermitian form h . If $\mathcal{E}$ and $\mathcal{F}$ are two ordered bases of M with change-of-basis matrix $P \in \text{GL}_n(A)$ (so $f_j = \sum_i e_i P(i, j)$), then

$$H_{\mathcal{F}} = P^* H_{\mathcal{E}} P,$$

where $H_{\mathcal{E}}$ and $H_{\mathcal{F}}$ are the matrices of h with respect to $\mathcal{E}$ and $\mathcal{F}$, respectively. Thus the two matrices are *-congruent.

Definition B.0.1. A *global field* is a field K that is either:

- a number field, i.e. a finite extension of $\mathbb{Q}$, or
- a global function field, i.e. a finite extension of the field of rational functions $\mathbb{F}_q(t)$ in one variable over a finite field $\mathbb{F}_q$.

Definition B.0.2. Let K be a global field (Definition B.0.1) and let v be a place of K . Write $|\cdot|_v$ for an absolute value representing v .

The *completion of K at v* , often denoted K_v , is the completion of K with respect to the metric induced by $|\cdot|_v$.

Definition B.0.3. Let $\{X_i\}_{i \in I}$ be a family of topological spaces indexed by a set I . For each $i \in I$, let $K_i \subseteq X_i$ be a topological subspace.

The *restricted product topology* on the restricted product

$$\prod'_{i \in I} X_i := \left\{ (x_i)_{i \in I} \in \prod_{i \in I} X_i \mid x_i \in K_i \text{ for all but finitely many } i \in I \right\},$$

with respect to the subsets $\{K_i\}_{i \in I}$, is the coarsest topology such that:

- The natural inclusion maps $X_j \rightarrow \prod'_{i \in I} X_i$, defined by $x_j \mapsto (y_i)$ where $y_j = x_j$ and $y_i = k_i$ (a fixed element in K_i) for all $i \neq j$, are continuous for all $j \in I$.
- The subspace topology on the product $\prod_{i \in F} X_i$ for any finite subset $F \subseteq I$ (where coordinates outside F are fixed in K_i) coincides with the product topology on finitely many factors.

Equivalently, the restricted product topology is generated by the base consisting of sets of the form

$$\prod_{i \in F} U_i \times \prod_{i \notin F} K_i,$$

where $F \subseteq I$ is finite, U_i are open sets in X_i , and outside F the coordinates lie in K_i .

Definition B.0.4. Let K be a global field (Definition B.0.1). Write M_K for the set of all places of K and write M_K^∞ for the set of archimedean places of K . Let $S \subseteq M_K$ be some subset of places of K (typically, S is a finite set). For each $v \in M_K$, write $\mathcal{O}_v$ for the ring of integers in the completion K_v (Definition B.0.2)

- The *adèle ring of K* , denoted $\mathbb{A}_K$, is the restricted direct product (Definition B.0.3) of the K_v (over all places v of K), with respect to the $\mathcal{O}_v$ at non-archimedean v :

$$\mathbb{A}_K = \left\{ (x_v)_v \in \prod_{v \in M_K} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *idèle group of K* , commonly denoted $\mathbb{A}_K^\times$ or $\mathbb{I}_K$, is the group of invertible elements of $\mathbb{A}_K$:

$$\mathbb{I}_K = \mathbb{A}_K^\times = \left\{ (x_v)_v \in \prod_{v \in M_K} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\},$$

where $\mathcal{O}_v^\times$ denotes the group of units of $\mathcal{O}_v$ for non-archimedean v .

- The *adèle ring outside S of K* , commonly denoted $\mathbb{A}_K^S$ or $\mathbb{A}_{K,S}$, is the restricted product of the completions K_v over all places $v \in M_K \setminus S$, with respect to the rings of integers $\mathcal{O}_v$ at non-archimedean places:

$$\mathbb{A}_{K,S} = \mathbb{A}_K^S = \left\{ (x_v)_v \in \prod_{v \in M_K \setminus S} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *idèle group outside S of K* , commonly denoted $(\mathbb{A}_K^\times)^S$, $(\mathbb{A}_{K,S}^\times)$, $\mathbb{I}_K^S$, or $\mathbb{I}_{K,S}$ is the group of invertible elements of $\mathbb{A}_K^S$:

$$(\mathbb{A}_K^\times)^S = \left\{ (x_v)_v \in \prod_{v \in M_K \setminus S} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *ring of finite adèles of K* , commonly denoted $\mathbb{A}_{K,\text{fin}}$, $\mathbb{A}_K^{\text{fin}}$, $\mathbb{A}_{K,\text{f}}$, $\mathbb{A}_K^{\text{f}}$, is the adèle ring outside $S = M_K^\infty$, the set of archimedean places of K :

$$\mathbb{A}_{K,\text{fin}} := \mathbb{A}_K^{M_K^\infty} = \left\{ (x_v)_v \in \prod_{v \notin M_K^\infty} K_v \mid x_v \in \mathcal{O}_v \text{ for all but finitely many non-archimedean } v \right\}.$$

- The *finite idèle group of K* , commonly denoted $\mathbb{A}_{K,\text{fin}}^\times$, $\mathbb{I}_{K,\text{fin}}$, $\mathbb{I}_K^{\text{fin}}$, $\mathbb{I}_{K,\text{f}}$, $\mathbb{I}_K^{\text{f}}$ etc. is the group of units of the ring of finite adèles:

$$\mathbb{A}_{K,\text{fin}}^\times := (\mathbb{A}_K^\times)^{M_K^\infty} = \left\{ (x_v)_v \in \prod_{v \notin M_K^\infty} K_v^\times \mid x_v \in \mathcal{O}_v^\times \text{ for all but finitely many non-archimedean } v \right\}.$$

All of these are equipped with the restricted product topology (Definition B.0.3) induced by the topologies of the local fields K_v and the subspace topologies thereof.

Definition B.0.5 (Adelic points of an algebraic group). Let K be a global field (Definition B.0.1) with ring of integers $\mathcal{O}_K$, and let G be a linear algebraic group (Definition B.0.7) defined over K .

- The *group of adelic points of G* is the group $G(\mathbb{A}_K)$ defined by

$$G(\mathbb{A}_K) := \prod'_v G(K_v),$$

(Definition B.0.2) where the restricted product is taken with respect to the compact open subgroups $G(\mathcal{O}_v)$ for almost all non-archimedean v .

- For a set S (usually finite) of places of K , the **group of adelic points of G outside S** is the group $G(\mathbb{A}_{K,S})$ defined by

$$G(\mathbb{A}_{K,S}) := \prod'_{v \notin S} G(K_v).$$

- The **group of finite adelic points** is the group $G(\mathbb{A}_{K,\text{fin}}) = G(\mathbb{A}_K^{\text{fin}})$ defined by

$$G(\mathbb{A}_K^{\text{fin}}) := \prod'_{v \text{ non-archimedean}} G(K_v).$$

- The **group of archimedean adelic points** is the group $G(K_\infty)$ defined by

$$G(K_\infty) := \prod_{v \text{ archimedean}} G(K_v).$$

Thus one has a natural factorization

$$G(\mathbb{A}_K) \cong G(\mathbb{A}_K^{\text{fin}}) \times G(K_\infty).$$

The groups $G(\mathbb{A}_K)$, $G(\mathbb{A}_{K,S})$, and $G(\mathbb{A}_K^{\text{fin}})$ are all topological groups (Definition A.0.1) under the restricted product topology (Definition B.0.3) induced by the topologies of the local fields K_v and the subspace topologies of $\mathcal{O}_v$ and become topological groups. The group $G(K_\infty)$ is a topological group under the direct product topology.

Definition B.0.6 (Height function on $G(\mathbb{A}_F)$). Let F be a global field (Definition B.0.1), and let $\mathbb{A}_F$ be the ring of adeles of F (Definition B.0.4). Let G be a linear algebraic group (Definition B.0.7) defined over F embedded as an algebraic subgroup in GL_n .

For each place v of F , let $\|\cdot\|_v$ be a norm on $G(F_v) \subseteq \text{GL}_n(F_v)$ defined as follows:

$$\|g_v\|_v := \max_{1 \leq i, j \leq n} |(g_v)_{ij}|_v,$$

(**♠ TODO: define the usual absolute value on F_v**) where $|\cdot|_v$ is the usual absolute value or valuation on F_v .

Then, for $g = (g_v)_v \in G(\mathbb{A}_F)$, define the adelic height function by the product

$$\|g\| := \prod_v \|g_v\|_v,$$

where the product converges due to $\|g_v\|_v = 1$ for all but finitely many non-archimedean places.

Definition B.0.7 (Linear algebraic group over a scheme). Let S be a base scheme. A **linear algebraic group over S** is an affine group scheme G over S that is finitely presented and smooth over S , and such that for some integer $n \geq 1$, there exists a closed immersion of S -group schemes

$$G \hookrightarrow \text{GL}_{n,S}.$$

B.1. Hecke characters and L -functions.

Definition B.1.1. (♠ TODO: TODO: can this be more generally defined for function fields?)

Let K be a global field. A *Hecke character* (or *Größencharacter*) of K is a character/continuous group homomorphism

$$\chi : \mathbb{A}_K^\times \rightarrow \mathbb{C}^\times$$

from the idèle group $\mathbb{A}_K^\times$ (Definition B.0.4) such that there exists an integer $m \geq 0$ (the ∞ -type) and a finite character χ_f of the finite idèles (Definition B.0.4) satisfying

$$\chi(x) = \chi_f(x)|x|^m$$

for all $x \in K^\times$ embedded diagonally in $\mathbb{A}_K^\times$, or more generally: χ is trivial on $K^\times$ (viewed as a subgroup of $\mathbb{A}_K^\times$ by the diagonal embedding). If $|\chi(a)| = 1$ for all $a \in \mathbb{A}_K^\times$, then χ is called a *unitary Hecke character*. (♠ TODO: TODO: define the weight of a hecke character)

Definition B.1.2. Let K be a number field and χ a Hecke character of K (Definition B.1.1). For $s \in \mathbb{C}$ with $\Re(s) > 1$, the *Hecke L -function of χ* is defined by the absolutely convergent sum

$$L(s, \chi) = \sum_{\mathfrak{a}} \chi(x_{\mathfrak{a}}) N(\mathfrak{a})^{-s}$$

where the sum runs over all nonzero ideals $\mathfrak{a}$ of $\mathcal{O}_K$, and $x_{\mathfrak{a}} = (\varpi_{\mathfrak{p}}^{n_{\mathfrak{p}}}) \in \mathbb{A}_{K,f}^\times$, where the prime factorization of $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$; this is well defined.

Equivalently, $L(s, \chi)$ may be defined by the absolutely convergent Euler product

$$L(s, \chi) = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\varpi_{\mathfrak{p}}) N_{K/\mathbb{Q}}(\mathfrak{p})^{-s}},$$

where the product runs over all nonzero prime ideals $\mathfrak{p}$ of $\mathcal{O}_K$, $\varpi_{\mathfrak{p}}$ is a uniformizer at $\mathfrak{p}$, and $N_{K/\mathbb{Q}}(\mathfrak{p}) = |\mathcal{O}_K/\mathfrak{p}|$.

Definition B.1.3 (Functional equation for Hecke L -functions). (♠ TODO: TODO: define the conductor of a hecke character) Let K be a number field with absolute discriminant $|\Delta_{K/\mathbb{Q}}|$, and let χ be a Hecke character of K (Definition B.1.1) with conductor $\mathfrak{f}(\chi)$. For $s \in \mathbb{C}$, define the *completed (Hecke) L -function* by

$$\Lambda(s, \chi) := (|\Delta_{K/\mathbb{Q}}| \cdot N(\mathfrak{f}(\chi)))^{\frac{s}{2}} \cdot \prod_{v|\infty} L_v(s, \chi) \cdot L(s, \chi),$$

where (♠ TODO: TODO: define the norm) (♠ TODO: TODO: define the Gamma function)

- $N(\mathfrak{f}(\chi))$ is the norm of the conductor ideal $\mathfrak{f}(\chi)$,
- the product runs over all infinite places v of K ,
- the *infinite local factors* $L_v(s, \chi)$ are defined using the Γ -function:

$$L_v(s, \chi) := \begin{cases} \Gamma_{\mathbb{R}}(s + \kappa_v) := \pi^{-\frac{s+\kappa_v}{2}} \Gamma\left(\frac{s+\kappa_v}{2}\right), & \text{if } v \text{ is a real place,} \\ \Gamma_{\mathbb{C}}(s + \kappa_v) := 2(2\pi)^{-(s+\kappa_v)} \Gamma(s + \kappa_v), & \text{if } v \text{ is a complex place,} \end{cases}$$

with some integers $\kappa_v \in \{0, 1\}$ determined by the ∞ -type of χ ,

- and $L(s, \chi)$ is the Hecke L -function (Definition B.1.2).

Theorem B.1.4 (Functional equation for Hecke L -functions). (♠ TODO: TODO: define the conductor of a hecke character) Let K be a number field with absolute discriminant $|\Delta_{K/\mathbb{Q}}|$, and let χ be a Hecke character of K (Definition B.1.1) with conductor $\mathfrak{f}(\chi)$. The completed Hecke L -function (Definition B.1.3) $\Lambda(s, \chi)$ satisfies the function equation

$$\Lambda(s, \chi) = \varepsilon(\chi) \Lambda(1 - s, \bar{\chi}),$$

(♠ TODO: TODO: define root number) (♠ TODO: Argue that the Hecke L -function is meromorphic; this function equation should say that the L -function is meromorphic for the real part of s greater than 1 or less than 0, but an argument for what happens in the critical strip is necessary.) where $\varepsilon(\chi) \in \mathbb{C}$ is the root number with $|\varepsilon(\chi)| = 1$, and $\bar{\chi}$ is the complex conjugate Hecke character.

Definition B.1.5 (Mellin transform). Let $f : \mathbb{R}_{>0} \rightarrow \mathbb{C}$ be a measurable function. The Mellin transform of f is the function $\mathcal{M}[f] : \mathbb{C} \rightarrow \mathbb{C}$ defined by

$$\mathcal{M}[f](s) := \int_0^\infty f(x) x^{s-1} dx,$$

whenever the integral exists (absolutely or as an improper integral).

Although this definition is stated for general measurable functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{C}$, it is intuitively meaningful when in the case that f is a function into $\mathbb{C}^\times$ — the function f would be a function from $\mathbb{R}_{>0}$, the identity component of the multiplicative group $\mathbb{R}^\times$, to the multiplicative group $\mathbb{C}^\times$. It is further intuitively meaningful to express the integral as

$$\int_0^\infty f(x) x^s \frac{dx}{x},$$

and regarding $\frac{dx}{x}$ as a Haar measure (♠ TODO: on what?)

Definition B.1.6. Let F be a global field (Definition B.0.1) and $\mathbb{A}_F$ its adele ring (Definition B.0.4). Write by $\mathbb{A}_F^\times$ the ideles (Definition B.0.4) of F , and let $\chi : \mathbb{A}_F^\times / F^\times \rightarrow \mathbb{C}^\times$ be a quasi-character. For $\varphi \in \mathcal{S}(\mathbb{A}_F)$ (the Schwartz–Bruhat space) and $s \in \mathbb{C}$ with $\Re(s)$ sufficiently large, *zeta integral/function of φ and χ* is defined by

$$Z(\varphi, \chi, s) = \int_{\mathbb{A}_F^\times} \varphi(x) \chi(x) |x|_{\mathbb{A}_F}^s d^\times x,$$

(♠ TODO: completed L -function) where $|\cdot|_{\mathbb{A}_F}$ is the idelic norm.

APPENDIX A. COMPLEX ANALYSIS

Theorem A.0.1. Let $U \subset \mathbb{C}$ be an open set and let $(f_n)_{n=1}^\infty$ be a sequence of holomorphic functions $f_n : U \rightarrow \mathbb{C}$. Suppose that (f_n) converges to a function $f : U \rightarrow \mathbb{C}$ uniformly on every compact subset of U . Then:

1. The limit function f is holomorphic on U .

2. For each $k = 0, 1, 2, \dots$, the sequence of derivatives $(f_n^{(k)})$ converges to $f^{(k)}$ uniformly on every compact subset of U .

Proof. (♠ TODO: AI generated, verify) We first establish the continuity of f , then prove that f is holomorphic, and finally prove the uniform convergence of derivatives on compact subsets.

We begin with the continuity of the limit function. Let $z_0 \in U$ be arbitrary. Since U is open, there exists a compact neighborhood K of z_0 contained in U (♠ TODO: state and reference a general fact that there is a compact subneighborhood of an open neighborhood — in particular, is this only true in Euclidean spaces or is it more true of other kinds of general topological spaces that euclidean spaces just happen to be an example of?). By the hypothesis of uniform convergence on compact subsets, (f_n) converges to f uniformly on K . Each f_n is holomorphic and hence continuous by ???. By ??, the uniform limit f on K is continuous at z_0 . Since z_0 was arbitrary, f is continuous on U .

Next we prove that f is holomorphic. Fix $z_0 \in U$ and choose $r > 0$ such that the closed disk $\overline{D}(z_0, r)$ is contained in U . Let γ be the positively oriented circle of radius r centered at z_0 , parametrized by $\gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$.

For each n , since f_n is holomorphic on U and the open disk $D(z_0, r)$ is simply connected, ??? applied to f_n on $D(z_0, r)$ and the curve γ gives, for every $z \in D(z_0, r)$,

$$f_n(z) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f_n(w)}{w - z} dw.$$

The image $\gamma([0, 2\pi])$ is compact, so by the hypothesis of uniform convergence on compact subsets, (f_n) converges uniformly to f on $\gamma([0, 2\pi])$. For each fixed $z \in D(z_0, r)$, the function $w \mapsto \frac{1}{w - z}$ is bounded on $\gamma([0, 2\pi])$. Hence,

$$\frac{f_n(\gamma(t))}{\gamma(t) - z} \gamma'(t) \longrightarrow \frac{f(\gamma(t))}{\gamma(t) - z} \gamma'(t)$$

uniformly on $[0, 2\pi]$ as $n \rightarrow \infty$.

By ??? applied separately to the real and imaginary parts of the integrand, the limit commutes with the integral. Therefore, for every $z \in D(z_0, r)$,

$$f(z) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{w - z} dw.$$

We now show that this integral representation implies that f is holomorphic on $D(z_0, r)$. For $z, z + h \in D(z_0, r)$ with $h \neq 0$, we compute

$$\begin{aligned} \frac{f(z + h) - f(z)}{h} &= \frac{1}{2\pi i} \oint_{\gamma} f(w) \cdot \frac{1}{h} \left(\frac{1}{w - z - h} - \frac{1}{w - z} \right) dw \\ &= \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z - h)(w - z)} dw. \end{aligned}$$

We compare this with the candidate derivative $\frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w-z)^2} dw$. The difference is

$$\frac{f(z+h) - f(z)}{h} - \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w-z)^2} dw = \frac{1}{2\pi i} \oint_{\gamma} f(w) \cdot \frac{h}{(w-z-h)(w-z)^2} dw.$$

For $|h|$ sufficiently small, specifically $|h| < \frac{r-|z-z_0|}{2}$, we have for all $w \in \gamma$ the estimates

$$|w-z| \geq r - |z-z_0|, \quad |w-z-h| \geq |w-z| - |h| \geq \frac{r-|z-z_0|}{2}.$$

Hence,

$$\begin{aligned} \left| \frac{f(z+h) - f(z)}{h} - \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w-z)^2} dw \right| &\leq \frac{|h|}{2\pi} \oint_{\gamma} \frac{|f(w)|}{|w-z-h||w-z|^2} |dw| \\ &\leq \frac{|h|}{2\pi} \cdot \sup_{w \in \gamma} |f(w)| \cdot \frac{1}{(r-|z-z_0|)^3} \cdot (2\pi r). \end{aligned}$$

The right-hand side tends to 0 as $h \rightarrow 0$. Consequently, the limit of the difference quotient exists and equals $\frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w-z)^2} dw$, i.e.,

$$f'(z) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w-z)^2} dw$$

for every $z \in D(z_0, r)$. Thus f is complex differentiable on $D(z_0, r)$, hence holomorphic on $D(z_0, r)$ by ???. Since z_0 was arbitrary, f is holomorphic on all of U .

Finally we prove the uniform convergence of derivatives on compact subsets. Let $K \subset U$ be an arbitrary compact set. For each $z \in K$, choose $r_z > 0$ such that $\overline{D}(z, 2r_z) \subset U$. The collection $\{D(z, r_z) : z \in K\}$ is an open cover of K , from which we extract a finite subcover $\{D(z_j, r_j)\}_{j=1}^m$.

For each j , let γ_j be the positively oriented circle of radius $2r_j$ centered at z_j , so that $\overline{D}(z_j, r_j) \subset D(z_j, 2r_j) \subset U$ and γ_j lies in U . By ??, for each n and each $z \in D(z_j, r_j)$ we have

$$f_n^{(k)}(z) = \frac{k!}{2\pi i} \oint_{\gamma_j} \frac{f_n(w)}{(w-z)^{k+1}} dw.$$

Since we have already shown that f is holomorphic on U , the same theorem applied to f yields

$$f^{(k)}(z) = \frac{k!}{2\pi i} \oint_{\gamma_j} \frac{f(w)}{(w-z)^{k+1}} dw.$$

For $z \in \overline{D}(z_j, r_j)$ and $w \in \gamma_j$, we have $|w - z| \geq |w - z_j| - |z - z_j| \geq 2r_j - r_j = r_j$. Therefore,

$$\begin{aligned} |f_n^{(k)}(z) - f^{(k)}(z)| &\leq \frac{k!}{2\pi} \oint_{\gamma_j} \frac{|f_n(w) - f(w)|}{|w - z|^{k+1}} |dw| \\ &\leq \frac{k!}{2\pi} \cdot \frac{\sup_{w \in \gamma_j} |f_n(w) - f(w)|}{r_j^{k+1}} \cdot (4\pi r_j) \\ &= \frac{2k!}{r_j^k} \sup_{w \in \gamma_j} |f_n(w) - f(w)|. \end{aligned}$$

Since $f_n \rightarrow f$ uniformly on the compact set γ_j , the right-hand side tends to 0 as $n \rightarrow \infty$, uniformly for $z \in \overline{D}(z_j, r_j)$ and hence for $z \in D(z_j, r_j)$. Taking the maximum over the finite index set $\{1, \dots, m\}$, we obtain

$$\sup_{z \in K} |f_n^{(k)}(z) - f^{(k)}(z)| \rightarrow 0$$

as $n \rightarrow \infty$. Thus $(f_n^{(k)})$ converges to $f^{(k)}$ uniformly on K , and since K was an arbitrary compact subset of U , the convergence is uniform on every compact subset of U . $\square$

Theorem A.0.2 (Lipschitz summation formula for reciprocal powers on the upper half-plane). (♠ TODO: AI generated, verify) Let $k \geq 2$ be an integer and let $z \in \mathbb{H}$ be a point of the upper half-plane (Definition 1.0.4). Then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(z+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m z},$$

where $e^{2\pi i m z}$ denotes the complex exponential function and $(k-1)!$ is the value of the Gamma function $\Gamma(k)$ at the integer k .

Proof. (♠ TODO: AI generated, verify) We prove the identity by computing the Fourier transform of the function $x \mapsto (x+z)^{-k}$ on $\mathbb{R}$ and applying ??.

Fix $z \in \mathbb{H}$ and write $z = a + iy$ with $a \in \mathbb{R}$ and $y > 0$. Define $f : \mathbb{R} \rightarrow \mathbb{C}$ by

$$f(x) = \frac{1}{(x+z)^k} = \frac{1}{(x+a+iy)^k}.$$

For $k \geq 2$, the function f is smooth and satisfies $|f(x)| \leq C|x|^{-k}$ for $|x| \gg 0$, hence $f \in L^1(\mathbb{R})$.

We compute the Fourier transform of f with respect to the communications and signal processing convention:

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} \frac{e^{-2\pi i \xi x}}{(x+z)^k} dx, \quad \xi \in \mathbb{R}.$$

Consider the contour integral over a large semicircle in the lower half-plane. For $\xi > 0$, the factor $e^{-2\pi i \xi x}$ decays as $\Im(x) \rightarrow -\infty$ (since $e^{-2\pi i \xi(u+iv)} = e^{-2\pi i \xi u} e^{2\pi \xi v}$, and for $v < 0$ this decays). For $\xi < 0$, the factor decays in the upper half-plane.

Case $\xi > 0$. Close the contour in the lower half-plane with a large semicircle of radius R . The integrand has a single pole of order k at $x = -z = -a - iy$, which lies in the lower half-plane because $\Im(-z) = -y < 0$. The contribution of the semicircle tends to 0 as $R \rightarrow \infty$ (since $k \geq 2$). By the residue theorem, the integral along the real axis equals $-2\pi i$ times the residue at $x = -z$ (the negative sign arises from the clockwise orientation of the closed contour in the lower half-plane).

The residue of $g(x) = e^{-2\pi i \xi x} (x+z)^{-k}$ at the pole $x = -z$ of order k is

$$\begin{aligned} \text{Res}(g, -z) &= \frac{1}{(k-1)!} \lim_{x \rightarrow -z} \frac{d^{k-1}}{dx^{k-1}} ((x+z)^k g(x)) \\ &= \frac{1}{(k-1)!} \lim_{x \rightarrow -z} \frac{d^{k-1}}{dx^{k-1}} e^{-2\pi i \xi x} \\ &= \frac{1}{(k-1)!} (-2\pi i \xi)^{k-1} e^{-2\pi i \xi(-z)} \\ &= \frac{(-2\pi i \xi)^{k-1}}{(k-1)!} e^{2\pi i \xi z}. \end{aligned}$$

Hence for $\xi > 0$,

$$\hat{f}(\xi) = -2\pi i \cdot \text{Res}(g, -z) = -2\pi i \cdot \frac{(-2\pi i \xi)^{k-1}}{(k-1)!} e^{2\pi i \xi z} = \frac{(-2\pi i)^k}{(k-1)!} \xi^{k-1} e^{2\pi i \xi z}.$$

Case $\xi < 0$. Close the contour in the upper half-plane. The function $g(x) = e^{-2\pi i \xi x} (x+z)^{-k}$ has no poles in the upper half-plane (the only pole is at $x = -z$, which lies in the lower half-plane). By the residue theorem, the contour integral encloses no poles, so

$$\hat{f}(\xi) = 0.$$

Case $\xi = 0$. The integral $\int_{-\infty}^{\infty} (x+z)^{-k} dx$ converges absolutely for $k \geq 2$ and equals 0 by a standard residue calculation (or by noting that the integral is unchanged under the change of variable $x \mapsto -x$ after shifting the contour).

Thus we have established

$$\hat{f}(\xi) = \begin{cases} \frac{(-2\pi i)^k}{(k-1)!} \xi^{k-1} e^{2\pi i \xi z}, & \xi > 0, \\ 0, & \xi \leq 0 \end{cases}$$

Application of Poisson summation. For $k \geq 2$, both f and $\hat{f}$ are in $L^1(\mathbb{R})$ (since $\hat{f}(\xi) \sim \xi^{k-1}$ near $\xi = 0$ and decays exponentially as $\xi \rightarrow \infty$). By the standard extension of Poisson summation to functions in $L^1(\mathbb{R})$ with L^1 Fourier transform, we have

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{m \in \mathbb{Z}} \hat{f}(m).$$

The left-hand side is

$$\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \frac{1}{(z+n)^k},$$

which converges absolutely for $k \geq 2$ by comparison with $\sum |n|^{-k}$.

The right-hand side is

$$\begin{aligned} \sum_{m \in \mathbb{Z}} \hat{f}(m) &= \hat{f}(0) + \sum_{m=1}^{\infty} \hat{f}(m) + \sum_{m=-\infty}^{-1} \hat{f}(m) \\ &= 0 + \sum_{m=1}^{\infty} \frac{(-2\pi i)^k}{(k-1)!} m^{k-1} e^{2\pi i m z} + 0 \\ &= \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m z}. \end{aligned}$$

Equating the two sides yields the claimed identity. $\square$

Theorem A.0.3 (Weierstrass M-test for uniform convergence of function series). Let $T = \tilde{R}_{\geq 0}$ be the positive cone of a totally ordered ring $\tilde{R}$ that is dense at zero. In particular, T is an ordered semiring with an adjoined infinity T^∞ . Let R be a commutative ring equipped with an absolute value valued in T . Let $(V, \|\cdot\|)$ be a complete extended normed module over R valued in T^∞ . Let S be a set and let $(f_n)_{n=1}^\infty$ be a sequence of functions $f_n : S \rightarrow V$.

Suppose there exists a sequence $(M_n)_{n=1}^\infty$ in $T_{\geq 0}$ such that:

1. For each n and each $x \in S$, $\|f_n(x)\| \leq M_n$;
2. The sequence (M_n) satisfies the Cauchy tails condition.

Then the series $\sum_{n=1}^\infty f_n(x)$ converges uniformly on S to a function $f : S \rightarrow V$.

In particular, since T is a subset of $\tilde{R}$, if the sequence of partial sums $(\sum_{n=1}^N M_n)_{N=1}^\infty$ converges in $\tilde{R}$, then (M_n) satisfies the Cauchy tails condition.

Proof. We prove pointwise convergence first, then uniform convergence.

Fix $x \in S$. The partial sums of the series $\sum f_n(x)$ form a sequence in V . For any $K > N$, by the triangle inequality for the norm, we have

$$\left\| \sum_{n=N+1}^K f_n(x) \right\| \leq \sum_{n=N+1}^K \|f_n(x)\| \leq \sum_{n=N+1}^K M_n.$$

By the Cauchy tails condition on (M_n) , for every $\varepsilon \in T_{>0}$ there exists N_0 such that for all $K > N \geq N_0$:

$$\sum_{n=N+1}^K M_n < \varepsilon.$$

Therefore, the partial sums satisfy, uniformly in x , the Cauchy condition:

$$\left\| \sum_{n=N+1}^K f_n(x) \right\| < \varepsilon \quad \text{for all } x \in S.$$

In other words, for any $K > N \geq N_0$, we have

$$\left\| \sum_{n=1}^K f_n(x) - \sum_{n=1}^N f_n(x) \right\| = \left\| \sum_{n=N+1}^K f_n(x) \right\| < \varepsilon,$$

i.e. the partial sums $(\sum_{n=1}^N f_n(x))_{N=1}^\infty$ form a Cauchy sequence in V ; recall that the extended metric on V is given by $d_V(v_1, v_2) = \|v_1 - v_2\|$. Since V is complete under this extended metric, for each fixed x the series $\sum f_n(x)$ thus converges to some limit, which we denote by $f(x)$. In fact, note that the convergence is uniformly Cauchy. By ??, since $(V, \|\cdot\|)$ is complete (with respect to the extended metric induced by $\|\cdot\|$), the convergence is thus uniform. $\square$

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